Cura-Link

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Faculty Of Computing
Riphah International University, Islamabad
Spring 2025

A Dissertation Submitted To

Faculty of Computing,

Riphah International University,
Islamabad

As a Partial Fulfillment of the Requirement for the Award of the Degree of

Bachelor of Science in Software Engineering

Faculty of Computing
Riphah International University, Islamabad

Date: May 20, 2025

Final Approval

This is to certify that we have read the report submitted by Usama Javed (25363), Muhammad Talha (25362) and Muhammad Nouman (27708) for the partial fulfillment of the requirements for the degree of the Bachelors of Science in Software Engineering (BSSE). It is our judgment that this report is of sufficient standard to warrant its acceptance by Riphah International University, Islamabad for the degree of Bachelors of Science in Software Engineering (BSSE).

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Declaration

We want to make it clear that our document named "Cura-Link" is entirely our own work, and we haven't copied it from anywhere. We've put in our personal efforts to complete this project along with the report. Our teachers and our supervisor Uzair Rasheed guided us throughout the process. We promise that none of the content in our system is taken from any other source or copied from another project. If it's proven otherwise, and any part is found to be a reproduction of someone else's work, we're ready to face the consequences. This declaration reflects our commitment to honesty and accountability in our academic endeavors.

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Dedication

Our heartfelt dedication reaches out to the pillars of our journey – our cherished family, steadfast friends, and dedicated teachers. Their unwavering encouragement and invaluable support have been the bedrock of our success in bringing "Cura-Link" to life. We extend sincere gratitude to our teachers for their exceptional support, guiding us with wisdom and care on our academic path. In genuine appreciation, we dedicate this work to our esteemed supervisor, "Mr. Uzair Rasheed". His unwavering commitment, continuous guidance, and transformative mentorship played a pivotal role in turning our ideas into a tangible reality. With deep thanks, we acknowledge the profound impact of these wonderful individuals on our journey. Their collective influence has not only shaped our success but has also added a rich dimension to our academic pursuits, making this endeavor all the more meaningful and fulfilling.

Acknowledgement

Foremost, we express our gratitude to Allah Almighty, the Merciful and the source of all knowledge, for endowing us with the courage and wisdom to bring this project to fruition. In His infinite mercy, we acknowledge the divine guidance that has been instrumental in our journey. Our heartfelt thanks extend to our family, friends, and teachers for their unwavering support.

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Abstract

Cura-Link is a mobile application which offers medical services like medicine ordering, nurse booking through bidding system and medical laboratory test booking. Cura-Link's Map feature makes sure every user gets their nearby services which will make their app experience better. Cura-Link also has an admin module through which admin will make sure all the service providers provide their services without any disappointments and misleading, admin will have the authority to blacklist any service provider violating Cura-Link's terms and conditions. Cura-Link ensures a reliable, patient-centered healthcare experience from booking to delivery.

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Chapter1:

Introduction

1 Introduction

1.1 Overview

The system is a mobile application which provides platform for medical patients to order medicines, book nurses and book lab test appointments at ease. System enables medical stores, nurses and medical labs to offer their services, it has features like a bidding system and an admin panel who is responsible for managing the service providers and ensuring users are getting services as promised.

1.2 Opportunities & Stakeholders

Below are opportunities and stakeholders explained.

1.2.1 Opportunities

The system offers opportunities by providing users with access to essential healthcare services such as medicine ordering, nurse booking and lab test booking, all in one platform. By enabling providers and users to bid for services, our application promotes competitive pricing and service customization, so that users can choose options that suit their needs in the best possible way. The system also supports local healthcare providers to reach out to more customers, and the admin panel ensures high quality service delivery by monitoring service provider's ratings and reliability, while the integrated map feature allows users to locate services in their preferred areas improving convenience overall.

1.2.2 Stakeholders

1. Patient
2. Medical Store Owner
3. Nurse
4. Laboratory Owner
5. Admin

1.3 Motivations and Challenges

Below are motivations and challenges explained.

1.3.1 Motivations for Cura-Link

1. **Usability:** System offers a user-friendly platform for medicine ordering, nurse booking and lab test booking making healthcare easier to use.
2. **Efficiency:** System's bidding system allows users to find and book healthcare services in comparatively less time, ensuring efficient healthcare delivery especially in urgent situations.
3. **Service Autonomy:** With features such as location-based services and bidding system, our app allows users to choose services according to their needs and budgets, ensuring personalized care for them.
4. **Cost-Effective:** System offers cost-effectiveness by introducing a bidding system in the application, enabling competition between service providers and that will create a cost-effective environment for users.

1.3.2 Challenges for Cura-Link

1. **Controlling Service Quality:** Ensuring that providers are providing quality and reliable services can be challenging, especially when expanding to new areas, so continuous monitoring and feedback mechanisms will be required.
2. **Competition:** As online healthcare services have become more common, System must differentiate itself or make itself better to make its mark on the market and for that continuous improvements will be important.
3. **Scope:** As the scope of cura-link is big, making sure we complete the app in given time is a crucial and challenging aspect in our project.
4. **Development Language/Environment:** As we never learned flutter(dart) in a course and cura link is being built on flutter so resolving complexities and learning flutter alongside is a great challenge for us.

1.4 Goals and Objectives

1. **Market Differentiator:** By providing bidding system for Nurse booking, our goal is to differentiate the system from the other platforms currently running in the market.
2. **Build Trust:** Our goal is to build trust between users and our platform by implementing reliability and admin monitoring, with that our application will attract more customers and trust will start to build.

3. **Growth:** Our aim is to maintain and continue the growth of our application with time so that system becomes one of the most used healthcare applications in Pakistan.

1.5 Solution Overview

We propose a solution which helps patients to access healthcare services by keeping in mind their choices, preferences, and budgets without compromising on the quality of services, also we enable service providers to increase their clientele and attract more traffic towards their businesses.

1.5.1 Scope of the Project

Below we have specified the complete scope for Cura-Link.

1.5.2 For Admin

1. Login.
2. Manage Service Providers(view,blacklist,whitelist).
3. Manage Profile.

1.5.3 For Patient

1. Register.
2. Verify email.
3. Login.
4. Reset Password.
5. Edit profile.
6. Order medicine.
7. Upload Prescription.
8. Call Nurse.
9. Accept nurse bid.
10. Cancel nurse booking request.
11. Book Laboratory test.
12. Ratings and feedback on services.

13. Online Payment.
14. Chat with service providers.
15. View Nurse Booking.
16. Filter nurse bookings.
17. Cancel nurse bookings.
18. Complete nurse bookings.
19. View Lab test booking.
20. Accept lab test booking.
21. Modify lab booking.
22. Cancel lab booking.

1.5.4 For Medical Stores

1. Register.
2. Verify email.
3. Login.
4. Reset Password.
5. Edit Profile.
6. View order requests.
7. Respond to order request
8. View pending orders.
9. View completed orders.
10. Chat with patient.
11. Verification.
12. Change availability

1.5.5 For Nurse

1. Register.

2. Verify email.
3. Login.
4. Reset Password.
5. Edit Profile.
6. View Booking requests.
7. Bid for booking requests.
8. View Bookings.
9. Filter bookings.
10. Complete booking.
11. Cancel booking.
12. Chat with patient.
13. Verification.
14. Change availability.

1.5.6 For Laboratories

1. Register.
2. Verify email.
3. Login.
4. Reset Password.
5. Edit Profile.
6. Chat with patient.
7. Verification.
8. View tests.
9. Add a test.
10. Delete a test.

11. Modify a test.

12. View Bookings.

13. Accept booking.

14. Modify booking.

15. Reject booking.

1.6 Chapter Summary

Chapter 1 explains the introductory part of our system, its opportunities, and stakeholders. The goals, motivations and challenges related to our project are also explained and after that we have given a short solution overview and enlisted Scope of the project.

Chapter2:
Market Survey

2 Chapter: Market Survey

In this chapter we have discussed the market survey we did.

2.1 Introduction

The idea behind market survey is to understand current applications and practices of market and increase our understanding of the system which are present and also how can we be better than them, and for that we conducted a market survey and interacted with people who were related to medical services, like Nurses, Medical Store owners and laboratory owners and gathered information from them through interviews which helped us a lot in understanding our requirements well and the flow of processes in this area of medical world.

Through our research and discussions with healthcare providers and patients, we found a strong need for a platform that brings all these services together in one place. With the system, we aim to simplify healthcare access by creating an app where patients can book services, view prices upfront, and choose providers that best match their needs. system also has a bidding option for flexibility in pricing and a map feature to help users find nearby services quickly. This new system will save time and give patients a smoother, more reliable way to get the care they need.

2.2 Literature Review

We studied current existing systems related to system and to compare their services with system we selected some of the important features from each system and , although some features were found common in all , some were unique, and some features weren't present at all which we wanted to implement in ours. Below is a table of comparison we performed.

Table 2.1: Market Survey Table

Applications on Market	Medicine Delivery	Nurse Booking	Lab test Booking	Bidding System	Service providers monitoring
Shifa4u	Yes	Yes	Yes	No	No
Ads Cares	No	Yes	Yes	No	No
E-Shifa	Yes	Yes	Yes	No	No
DVAGO	Yes	No	Yes	No	No

2.3 Chapter Summary

In this chapter of the Market Survey, we have described what we did in our survey and how it helped us clear up our doubts about our project. We have also compared existing systems and their functionalities , also we added services which we are offering, and others aren't so that the difference becomes clear.

Chapter 3

Requirement Engineering

3 Chapter: Requirement Engineering

In this chapter we have done requirements engineering for Cura-Link,

3.1 Introduction

In this chapter we will perform requirement, implement its processes and discuss about the Functional requirements of our project 'Cura-Link'. We will also discuss the problem statements we have found during research into our project idea. The functional requirements are gathered by techniques like brainstorming and interviews with relevant people.

3.2 Problem Scenarios

Table 3.1: Problem Statement-1

3.2.1 Problem Statement: Medicine Delivery	
The problem of	Patients struggle to get timely medicine delivery, especially at urgent times.
Affects	Patients in need of urgent medicines.
The results of which	Delays in medicine delivery and complications in patient health.
Benefits of	Patients can order medicines and get them in less time comparing to others.

Table 3.2: Problem Statement 2

3.2.2 Problem Statement: Service Price flexibility	
The problem of	Lack of autonomy in choosing cost of services.
Affects	Patients
The result of which	Patients may overpay or give up orders due to not being able to afford.
Benefits of	Cura-Link bidding feature will allow users to choose nurses costs according to their budget and preferences.

Table 3.3: Problem Statement 3

3.2.3 Problem Statement: Verified Service Providers	
The problem of	Difficulty in finding verified service providers for the healthcare needs of patients.
Affects	Patients having concerns over safety and service quality.
The result of which	Patients can have low service quality and scams.
Benefits of	Cura-Link offers verified service providers with ratings allowing patients to choose them with confidence.

Table 3.4: Problem Statement 4

3.2.4 Problem Statement: Admin Monitoring	
The problem of	Lack of monitoring over service provider quality and customer feedback.
Affects	Service quality and patient satisfaction.
The result of which	Service providers with low quality may continue work affecting patient's trust.
Benefits of	Cura-Link's admin monitoring will ensure better service delivery and blacklisting of those who break patient's trust.

Table 3.5: Problem Statement 5

3.2.5 Problem Statement: Coverage and Accessibility	
The problem of	Limited healthcare options in certain areas.

Affects	Overall service quality and patient satisfaction.
The result of which	Patients will have limited access to Cura-Link's services.
Benefits of	Cura-Link's map feature will allow users to find nearby service providers enhancing service delivery throughout an area.

3.3 Functional Requirements

Below we have complete functional requirements for Cura-Link.

3.3.1 Patient

Table 3.6: Patient's Functional Requirements

Requirement ID	Description
FRP-01	Patient will be able to register with Cura-Link.
FRP-02	Patient will be able to verify their email for registration.
FRP-03	Patient will be able to login to their account.
FRP-04	Patient will be able to reset his password.
FRP-05	Patient will be able to edit their profile.
FRP-06	Patient will be able to pay online/cod.
FRP-07	Patient will be able to chat with service providers.
FRP-08	Patient will be able to rate and give feedback to service providers.
FRP-09	Patient will be able to order medicine.
FRP-10	Patient will be able to attach prescription with medicine order.
FRP-11	Patient will be able to book Lab test from a laboratory.
FRP-12	Patient will be able to book nurse.
FRP-13	Patient will be able to accept a nurse's bid.
FRP-14	Patient will be able to cancel the booking request.

FRP-15	Patient will be able to view nurse booking.
FRP-16	Patient will be able to filter bookings.
FRP-17	Patient will be able to cancel the booking.
FRP-18	Patient will be able to complete booking.
FRP-19	Patient will be able to view lab bookings.
FRP-20	Patient will be able to accept lab bookings.
FRP-21	Patient will be able to modify lab bookings.
FRP-22	Patient will be able to cancel lab bookings.

3.3.2 Nurse

Table 3.7: Nurse's Functional Requirements

Requirement ID	**Description**
FRN-01	Nurse will be able to register with Cura-Link.
FRN-02	Nurse will be able to verify their email for registration.
FRN-03	Nurse will be able to login to their account.
FRN-04	Nurse will be able to reset their password.
FRN-05	Nurse will be able to Edit their profile.
FRN-06	Nurse will be able to chat with patient.
FRN-07	Nurse will be able to verify themselves through cura-link.
FRN-08	Nurse will be able to view booking requests.
FRN-09	Nurse will be able to Bid for booking request.
FRN-10	Nurse will be able to change their availability.
FRN-11	Nurse will be able to view bookings.
FRN-12	Nurse will be able to filter bookings.
FRN-13	Nurse will be able to complete booking.

FRN-14	Nurse will be able to cancel bookings.

3.3.3 Medical Store

Table 3.8: Medical Store's Functional Requirements

Requirement ID	Description
FRMS-01	Medical store will be able to register themselves with Cura-Link.
FRMS-02	Medical store will be able to verify their email for registration.
FRMS-03	Medical store will be able to log in to their account.
FRMS-04	Medical store will be able to reset their password.
FRMS-05	Medical store will be able to edit profile.
FRMS-06	Medical store will be able to chat with patient.
FRMS-07	Medical store will be able to verify through cura-link.
FRMS-08	Medical store will be able to change his/her availability.
FRMS-09	Medical store will be able to view order request.
FRMS-10	Medical store will be able to respond to order request.
FRMS-11	Medical store will be able to view pending orders.
FRMS-12	Medical store will be able to view completed orders.

3.3.4 Laboratory

Table 3.9: Lab's Functional Requirements

Requirement ID	Description
FRL-01	Lab will be able to register with Cura-Link.
FRL-02	Lab will be able to verify their email for registration.

FRL-03	Lab will be able to log in to their account.
FRL-04	Lab will be able to reset their password.
FRL-05	Lab will be able to edit profile.
FRL-06	Lab will be able to verify themselves through cura-link.
FRL-07	Lab will be able to chat with patient.
FRL-08	Lab will be able to view their test list.
FRL-09	Lab will be able to add a test.
FRL-10	Lab will be able to delete a test.
FRL-11	Lab will be able to edit a test.
FRL-12	Lab will be able to view bookings.
FRL-13	Lab will be able to accept a booking.
FRL-14	Lab will be able to modify booking.
FRL-15	Lab will be able to reject a booking.

3.3.5 Admin

Table 3.10: Admin's Functional Requirements

Requirement ID	**Description**
FRA-01	Admin will be able to login.
FRA-02	Admin will be able to manage service providers(blacklist/Whitelist).
FRA-03	Admin will be able to view users list.
FRA-04	Admin will be able to Edit his profile.

3.4 SQA Activities: Defect Detection

Below we have done defect detection for Cura-Link.

3.4.1 Separate Dashboards

Table 3.11: Defect detection- Separate Dashboard

Requirement	Checklist Point	Defect
Each stakeholder (Patient, Nurse, Medical Store, Medical Lab, Admin) has a separate dashboard.	Are the functionalities for each dashboard (e.g., Nurse bidding management, lab availability updates) clearly specified?	Lack of details about the specific features available on each stakeholder's dashboard.

3.4.2 Registration

Table 3.12: Defect Detection- Registration

Requirement	Checklist Point	Defect
All users (Patients, Nurses, Medical Stores, Medical Labs) can register an account.	Are all inputs like name, email, phone, and password specified, including formats and validation?	Missing input validation details for fields like phone number and password, including format specifications.

3.4.3 Notifications

Table 3.13: Defect Detection- Notifications

Requirement	Checklist Point	Defect
Users (Patients, Nurses, Medical Stores, Medical Labs) receive	Are the types of notifications (e.g., booking confirmation, delivery status, payment	Missing details about notification triggers (e.g., booking status changes)

notifications about their bookings or job status.	alerts) and their triggers specified?	and the types of notifications sent.

3.4.4 Real-time Communication (In-App Chat)

Table 3.14: Defect Detection- Chatting

Requirement	Checklist Point	Defect
Users (Patients, Nurses, Medical Store Owners) can communicate via in-app chat.	Is the in-app chat feature well-defined (e.g., message sending, receiving, notifications, error handling)?	Missing details about how the system handles message delivery failures or errors (e.g., network issues).

3.4.5 Service Provider Verification

Table 3.15: Defect Detection - Verification

Requirement	Checklist Point	Defect
Admin can verify service providers (e.g., nurses, medical stores, labs).	Are the steps for verifying service providers (e.g., verification of documents, user reviews) specified clearly?	Lack of clarity in the verification process, including criteria for verifying service providers and how the system handles failed verification attempts.

3.4.6 Error Handling

Table 3.16: Defect Detection- Error Handling

Requirement	Checklist Point	Defect
The system should provide meaningful error messages and log critical system failures for admin review.	Are error messages and system logs clearly defined (e.g., types of errors, visibility of errors to users, logging for admin)?	Missing definitions of error messages for common issues and lack of logging mechanisms for system crashes or unexpected behavior.

Chapter 4

System Design

4 Chapter: System Design

Below chapter is related to the complete system design of Cura-Link.

4.1 Introduction

System Design is an important aspect of software engineering to show something that is intended to be constructed. It is constructed by keeping in mind the requirements of our project. Without proper design there is a big risk of developing an unstable system that may fail at some point and its testing will become hard. In this chapter we will explore our system design which will include the architecture design, use case design and sequence diagram of Cura-Link.

4.2 Architectural Design

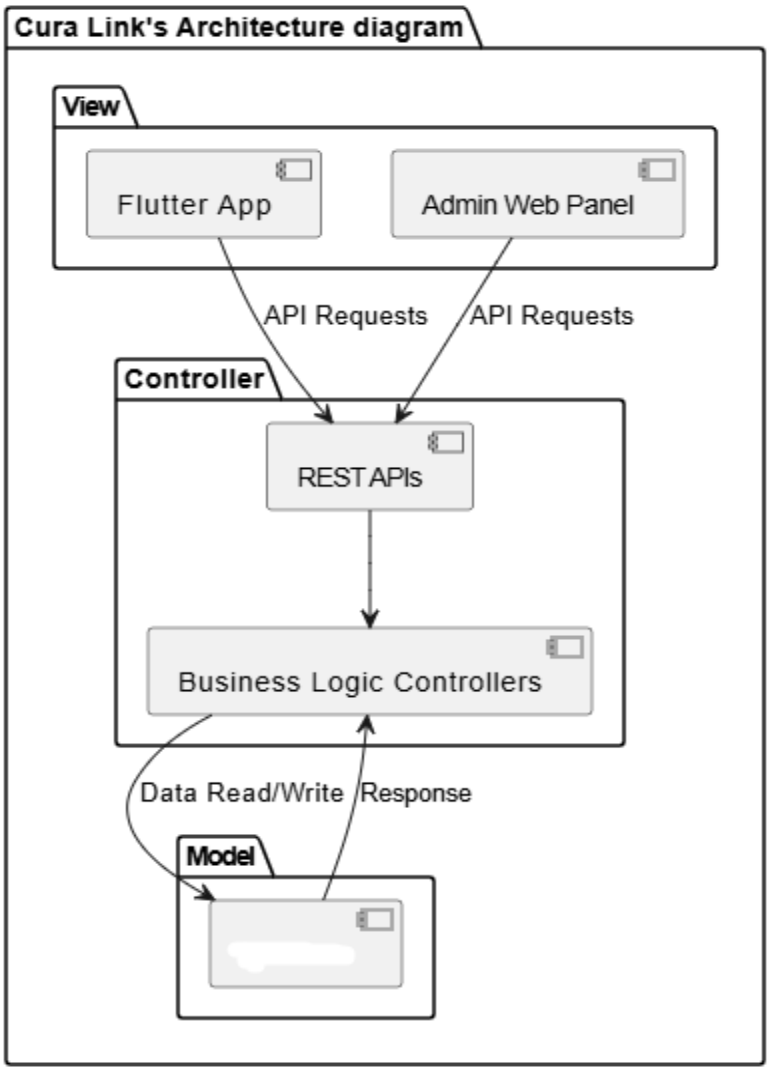

Figure 4.1: System Architecture Diagram

4.3 Use Case Diagrams

Below we have use case diagrams.

4.3.1 System Use Case Diagram

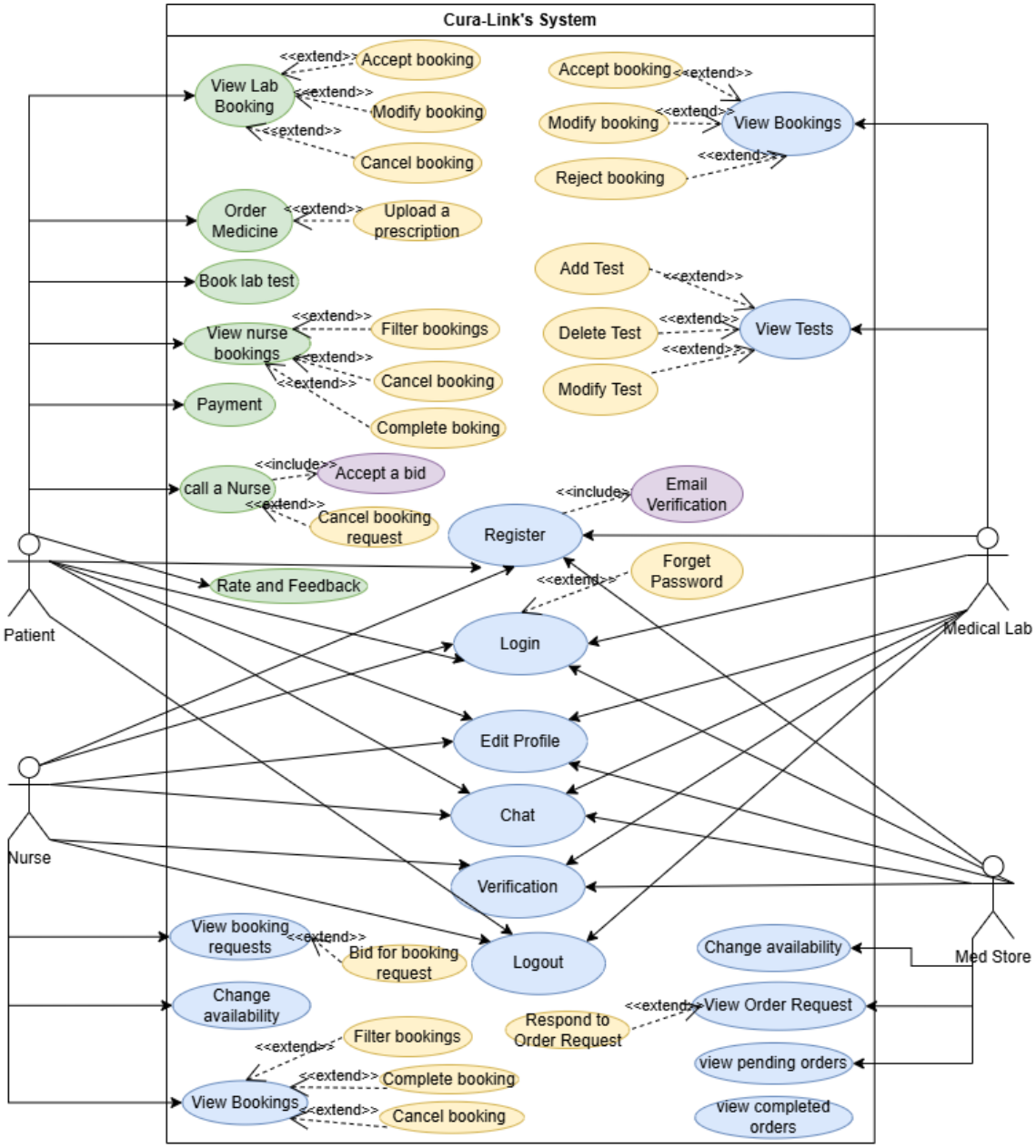

Figure 4.2: System Use Case Diagram

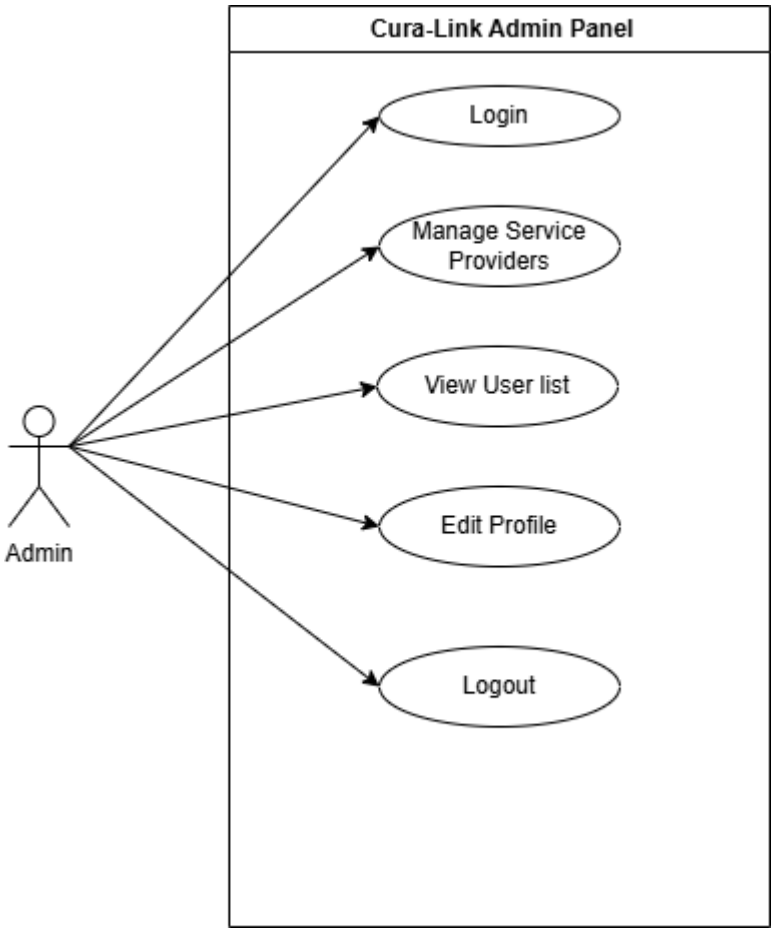

Figure 4.3: Admin Use Case Diagram

4.3.2 Patient Use Case Diagram

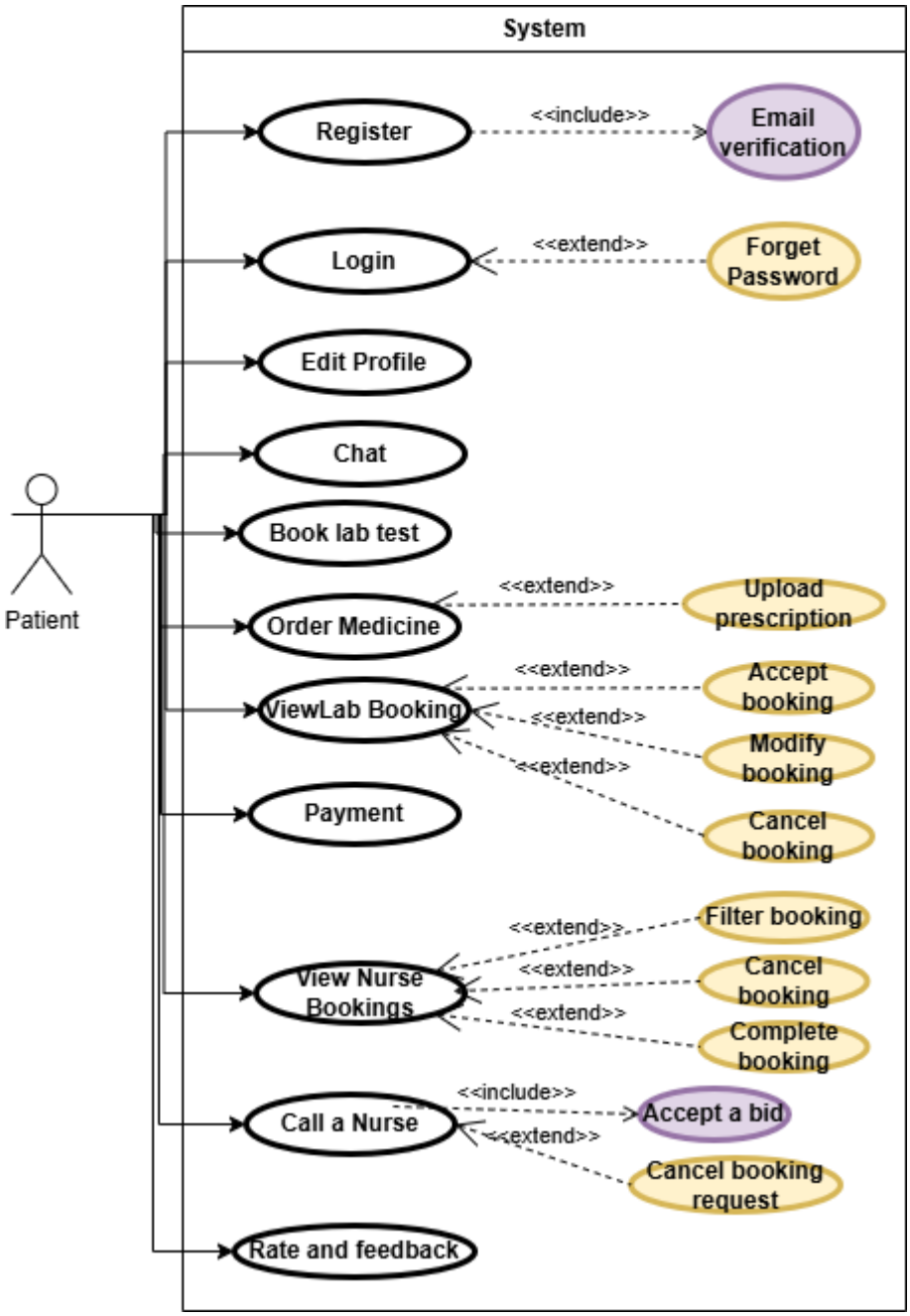

Figure 4.4: Patient Use Case Diagram

4.3.3 Nurse Use Case Diagram

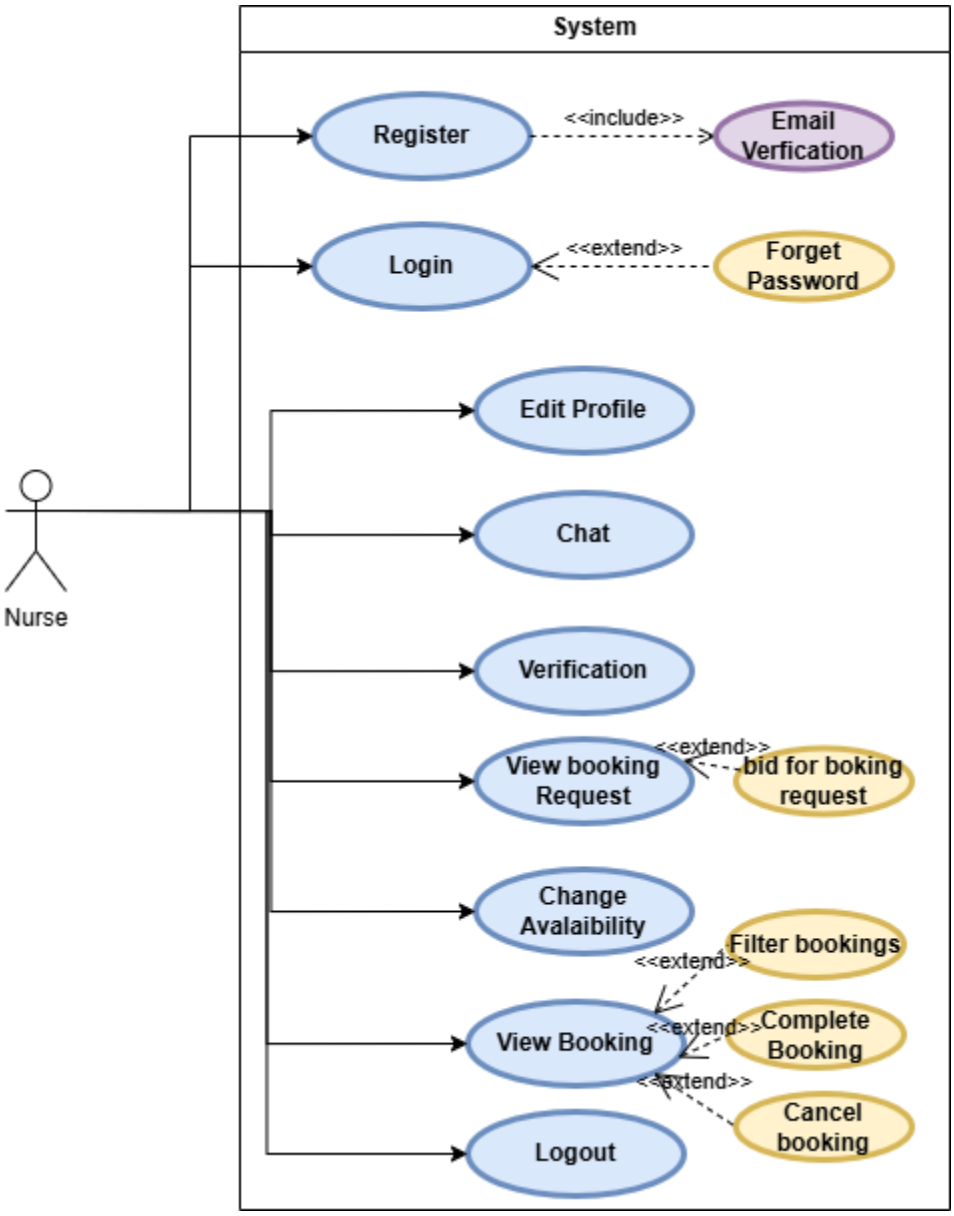

Figure 4.5: Nurse Use Case Diagram

4.3.4 Laboratory Use Case Diagram

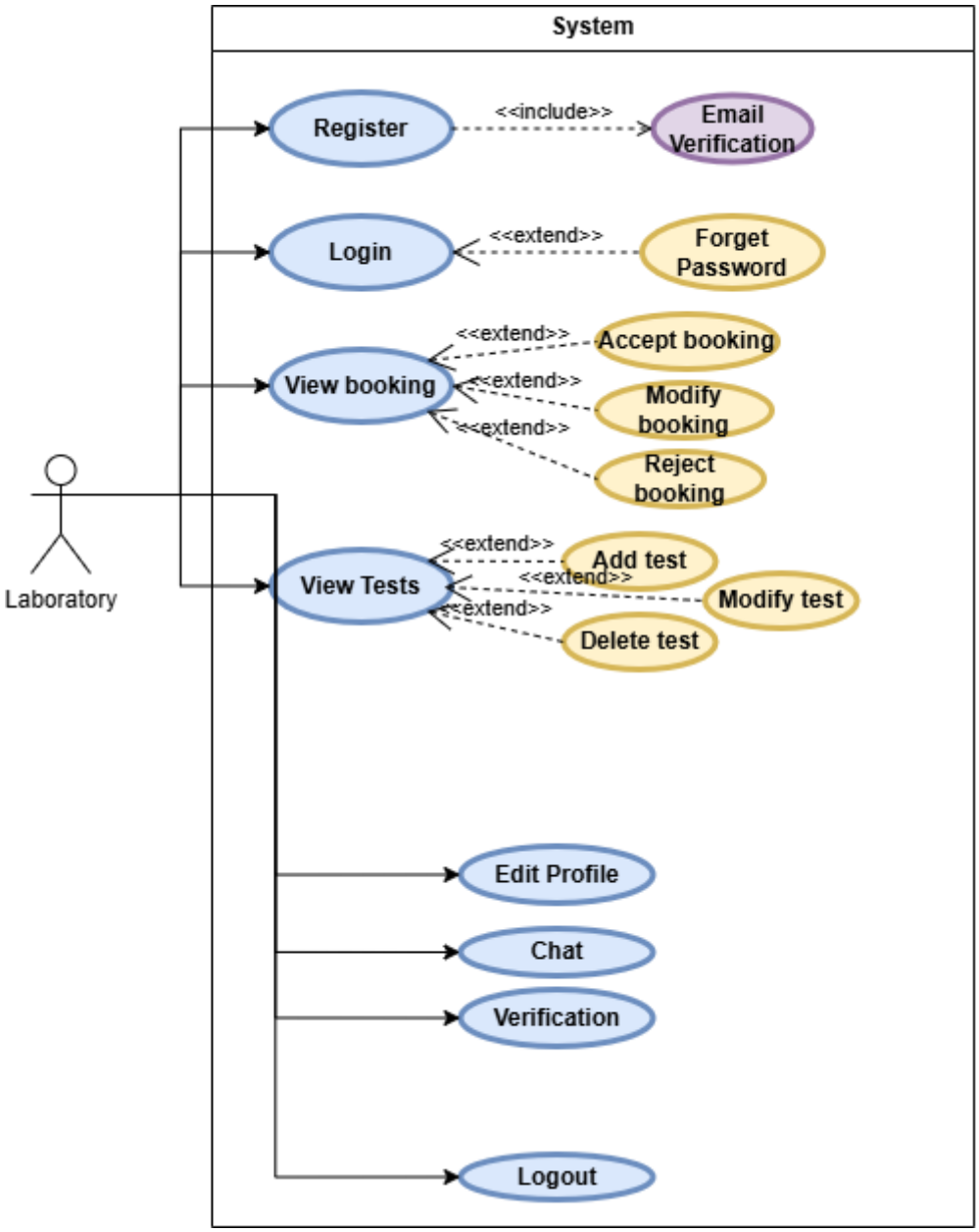

Figure 4.6: Lab Use Case Diagram

4.3.5 Medical Store Use Case Diagram

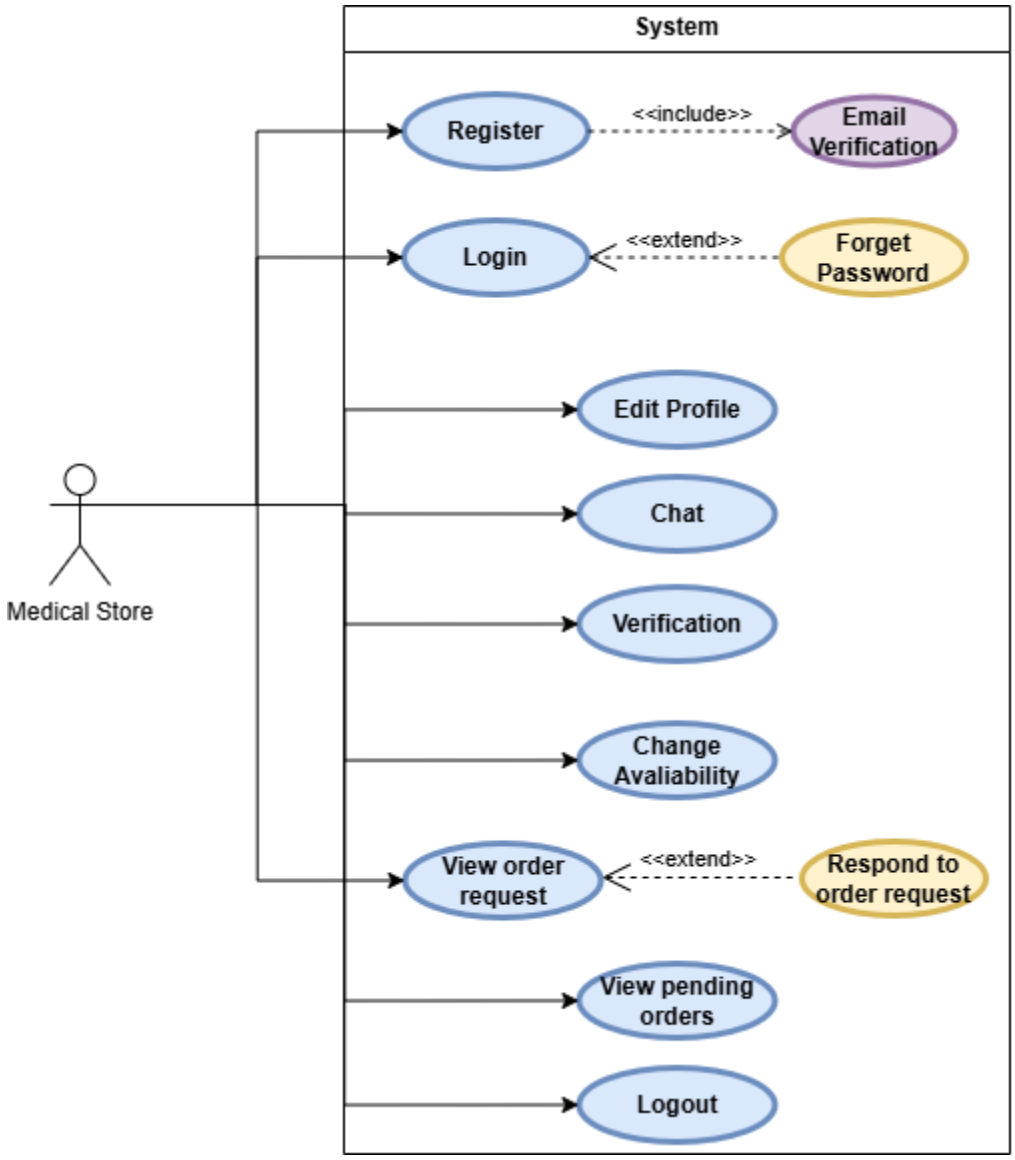

Figure 4.7: Medical Store Use Case Diagram

4.4 Fully Dressed Use Cases

Below we have generalized fully dressed use cases.

4.4.1 Generalized Use Cases

Generalized Use cases first.

4.4.1.1 Registration

Table 4.1: Registration FD Uc

UC-ID	GUC-01

Use Case Name	Registration.
Primary Actor	Nurse, Patient, Medical Store, Medical Lab
Purpose	To allow users (Nurse, Patient, Medical Store, Medical Lab) to register with Cura-Link.
Pre-Condition	User must have internet and cura link open.

Basic Flow	
User Action	**System Response**
1: Selects their role (e.g., Nurse, Patient, Medical Store, Medical Lab). **3:** Enters the required fields (e.g., name, phone number, email, and password). **5:** Verifies their account.	**2:** Displays the registration screen asking user to enter fields. **4:** Validates the input and sends an email/OTP to the user's provided contact details. **6:** Creates a new profile for the user and confirms the registration.

Alternative Flow
3A: If user's number or email already registered, system will ask the user to try with a different number or email. **3B:** If a user leaves any field empty system will ask user to enter all required fields.

Post-Condition	User is registered with Cura-Link. User is redirected to his dashboard.

Includes	
Use Case	**Condition**

Email Verification	User has provided valid details including valid email address.

4.4.1.2 Email Verification

Table 4.2: Email Verification FD

UC-ID	GUC-02
Use Case Name	Email verification.
Primary Actor	Nurse, Patient, Medical Store, Medical Lab
Purpose	To allow users (Nurse, Patient, Medical Store Owner, Medical Lab Owner) to verify their provided email for registration.
Pre-Condition	1. User is in the registration page. 2. User must have provided valid info in the registration page.
Basic Flow	
User Action	**System Response**
1: Provides information in the required fields in registration. **3:** Goes to his mail and verifies through the email user received. **5:** Goes back to app's registration screen.	**2:** Sends verification email and asks the user to verify their email through link in the mail. **4:** Displays verification confirmation and ask user to go back to app. **6:** Register user with cura link and redirect him to his dashboard.
Alternative Flow	
5A: If user goes back to app from mail without verifying, system will keep asking user to verify email.	

Post-Condition	User is logged in with Cura-Link. User is redirected to their respective dashboard.

4.4.1.3 Login

Table 4.3: Login FD Use case

UC-ID	GUC-03
Use Case Name	Login.
Primary Actor	Nurse, Patient, Medical Store, Medical Lab
Purpose	To allow users (Nurse, Patient, Medical Store Owner, Medical Lab Owner) to log in to Cura-Link.
Pre-Condition	User must be registered with Cura-Link.
Basic Flow	
User Action	**System Response**
1: Navigates to the login screen. **3:** Enters the required fields (e.g., email/phone number and password).	**2:** Displays the login screen with required fields. **4:** Verifies patient's credentials. **5:** Upon successful validation, the system redirects the user to their respective dashboard.
Alternative Flow	
3A: If the user's credentials doesn't match, system will ask user to re-enter them. **3B:** If user leaves any field empty system will ask user to enter all required fields.	
Post-Condition	User is logged in with Cura-Link. User is redirected to their respective dashboard.

Extension	
Use Case	**Condition**
Forget Password	User has clicked on forget password from login page.

4.4.1.4 Forget Password

Table 4.4: Forget Password FD

UC-ID	GUC-03-A
Use Case Name	Forget Password (Password Reset).
Primary Actor	Users (Patient, Nurse, Medical Store, Medical Lab)
Purpose	To allow users (Patient, Nurse, Medical Store, Medical Lab) reset their password.
Pre-Condition	User email must be registered.
Basic Flow	
User Action	**System Response**
1: Clicks on "Forget Password?" button from login page. **3:** Enters his email. **5:** Goes to mail and changes password through email.	**2:** Displays forget password section, ask for email id. **4:** Sends password reset link to user's email. **6:** Updates password and **s**hows "Password changed successfully, please head back to cura-link and login to your account".
Alternative Flow	
3A: If the user enters an unregistered email address. The system will notify the user that the email is not registered, and prompt them to enter a valid registered email.	
Post-Condition	User has changed his password successfully.

4.4.1.5 Logout

Table 4.5: Logout FD Use case

UC-ID	GUC-04
Use Case Name	Logout
Primary Actor	Users (Patient, Nurse, Medical Store, Medical Lab)
Purpose	To allow users(Patient, Nurse, Medical Store, Medical Lab) logout from cura-link.
Pre-Condition	Users must be logged in.
Basic Flow	
User Action	**System Response**
1: Navigates to his profile. **3:** Clicks on logout button.	**2:** Displays his profile with options. **4:** Logs out user from his account and redirect him to login screen.
Alternative Flow	
1A: If the system encounters a network issue during logout: The system displays an error message: "Logout failed. Please check your connection and try again.", The user remains logged in.	
Post-Condition	1. The user is securely logged out of the application. 2. The user's session is terminated.

4.4.1.6 Chat

Table 4.6: Chat FD Use case

UC-ID	GUC-05
Use Case Name	Chat
Primary Actor	Users (Patient, Nurse, Medical Store, Medical Lab)
Purpose	To allow communication between patients and service providers (Nurse, Medical Store, Medical Lab) regarding active or completed bookings/orders.
Pre-Condition	There must be an active or completed booking/order between the patient and the service provider.
Basic Flow	
User Action	**System Response**
1: Navigates to the "Chat" section or accesses it from an order/booking detail page. **3:** User sends a message. **5:** Other user receives the message and responds.	**2:** Displays the chat interface between the patient and the selected service provider. **4:** Delivers the message to the other user in real-time. **6:** Updates the conversation in real-time for both users.
Alternative Flow	
4A: If the user sends a message but there's a network issue, the system notifies the user with: "Message failed to send. Please try again".	
Post-Condition	Users (patients and service providers) have successfully interacted through the Cura-Link chat feature.

4.4.1.7 Verification

Table 4.7: Verification FD Use case

UC-ID	GUC-06
Use Case Name	Service provider verification.
Primary Actor	Medical Laboratory, Nurse, Medical lab
Purpose	To allow service providers (Nurse, Medical Store, Medical Lab) to verify their service authenticity using the system.
Pre-Condition	The service provider must be logged in.
Basic Flow	
Service provider Action	**System Response**
1: Clicks on the "Verify" button from the verification warning on the top of his dashboard. **3:** Enters the required info (e.g., name, ID/license number) and proceeds.	**2:** Redirects the service provider to the verification section. **4:** Compares the data with a dummy table and notifies the service provider about the result of their verification request.
Alternative Flow	
4A: if provided information isn't found in database, system will ask user that verification failed and retry with a valid information next time.	
Post-Condition	The service provider is verified and can now access verified-only features.

4.4.1.8 Edit Profile

Table 4.8: Manage Profile FD Use case

UC-ID	GUC-07
Use Case Name	Edit Profile.
Primary Actor	Patient, Nurse, Medical Store, Medical Lab, Patient.
Purpose	To allow service providers (Nurse, Medical Store, Medical Lab) to manage and update their account profiles.
Pre-Condition	User must be logged in.
Basic Flow	
User Action	**System Response**
1: Navigates to the profile section.	**2:** Displays user profile related to their role.
3: Clicks on edit profile.	**4:** Shows editable information set to be edited.
5: Updates the profile info and clicks "Save."	**6:** Updates the profile with the new changes.
Alternative Flow	
5A: If the User leaves the page without saving system will save revert last saved changes. **5B:** If any change isn't acceptable by the system , it should prompt user to try a different change.	
Post-Condition	The user's profile has been updated successfully.

4.4.2 Specific Use Cases

4.4.2.1 Order Medicine

Table 4.9: Order Med FD Use case

UC-ID	P01
Use Case Name	Order Medicine
Primary Actor	Patient
Purpose	To allow the patient to order medicines.
Pre-Condition	Patient must be logged in to his account.
Basic Flow	
Patient Action	**System Response**
1:Navigates to order medicine section. **3.** Search / Select medicines/upload prescription and click on checkout. **5:** Click on confirm. **7.** Enter the Delivery address and click on submit.	**2.** Display medicine list with option to attach precipitation. **4.** Display order summary with options to confirm or cancel. **6:**Redirects the patient to request creation screen and ask for delivery information. 8. Create the request notify stores and take patient to store offer screen.
Alternative Flow	
3A: If a patient proceeds without mentioning a medicine nor uploading a prescription system will prompt user to select a medicine or upload prescription. **4A:** If patient click on cancel systems cancel the request and redirect patient to dashboard. **7A**: if patients click on submit without entering delivery address system will ask the patient to enter the delivery address first.	
Post-Condition	The patient successfully ordered medicines.

4.4.2.2 Upload Prescription

Table 4.10: Upload Prescription FD Uc

UC-ID	P02
Use Case Name	Attach a Medicine Prescription.
Primary Actor	Patient
Purpose	To allow patient Attach a prescription while ordering for medicines.
Pre-Condition	Patient must have initialized order medicine flow.
Basic Flow	
Patient Action	**System Response**
1: Clicks on upload a prescription button during medicine ordering. **3:** Will browse and select prescription file(jpg,png) from device and click on upload.	**2:** Shows prescription upload option. **4:** Adds the prescription to patient's order and allow the patient to continue.
Alternative Flow	
3A: If patient select any other format file system will ask the user to upload a jpg or png file.	
Post-Condition	Patient has uploaded a medicine prescription.

4.4.2.3 Nurse Booking

Table 4.11: Nurse Booking FD Use case

UC-ID	P03
Use Case Name	Book Nurse.
Primary Actor	Patient
Purpose	To allow patient to book nurse.
Pre-Condition	Patient must be logged in to his account.
Basic Flow	
Patient Action	**System Response**
1: Navigates to "Call nurse" section.	**2:** Shows nursing services screen with available services and their prices.
3: Search/Select any service.	**4:** Creates the request notify nurses and take users to available bid screen.
5: Waits for bids from nurses.	**6:** Nurses will bid for the request and system will show the bids to patient.
7: Views the bids and selects preferred bid to confirm.	**8:** Confirms the booking and notifies nurse and labels the request as booked.
Alternative Flow	
7A: if user click on cancel booking request system will cancel booking request and delete it from the database and redirect it to dashboard.	
Post-Condition	Patient successfully make a nurse booking.

4.4.2.4 Rate & Feedback

Table 4.12: Rate & Feedback FD Uc

UC-ID	P04
Use Case Name	Rating & Feedback
Primary Actor	Patient
Purpose	To allow patient rate any service and give feedback about it.
Pre-Condition	Patient must complete any service order.
Basic Flow	
Patient Action	**System Response**
1: View a completed order. **3:** Clicks on rate button. **5:** Rate the service (1-5 stars) and write feedback and post it.	**2:** Shows order list with rate button on them. **4:** Shows rating dialogue with stars and feedback box. **6:** Records and save data.
Alternative Flow	
3A: Patient cant rate service before completing/Accepting Service. **5A:** If User submit Feedback without rating it shows warning message "Empty feedback can't be Saved".	
Post-Condition	Patient rates a service and gives feedback.

4.4.2.5 Lab Test Booking

Table 4.13: Lab Test Booking FD Use case

UC-ID	P05
Use Case Name	Lab Test Booking.
Primary Actor	Patient.
Purpose	To allow patient book lab test from medical labs.
Pre-Condition	The patient must be logged in.

Basic Flow	
Patient Action	**System Response**
1: Click on book lab. **3: S**elect specific lab. **5: C**lick on book appointment button. **7: S**elect any lab test service. **9:** Select date and time for booking and proceed.	**2:** Shows map with nearby labs in it. **4:** Shows lab related details with book appointment button. **6:** Displays all lab services with their price tags. **8**: Date&Time Picker shows to select date and time. **10**: Creates booking request and notify the lab.
Alternative Flow	
2A: If no labs are available, system notifies user that “no labs available nearby”.	
Post-Condition	Patient books a laboratory test slot successfully.

4.4.2.6 View Orders(Patient)

Table 4.14: View Orders FD Uc

UC-ID	P06
Use Case Name	View Orders
Primary Actor	Patient.
Purpose	To allow patient view his orders.
Pre-Condition	Patient has made an order.
Basic Flow	
Patient Action	**System Response**

1: Navigates to the "Mybooking" Section. **3:** Scrolls through orders and select an order.	**2:** Displays orders page with list of patient's medicine orderings. **4:** Shows selected order details.
Alternative Flow	
2A: If no orders found system will show empty orders page saying No booking Found.	
Post-Condition	Patient views his orders.

4.4.2.7 View Lab Booking(Patient)

Table 4.15: Manage Lab Booking FD Uc

UC-ID	P07
Use Case Name	Manage Lab Bookings.
Primary Actor	Patient.
Purpose	To allow patient manage lab booking.
Pre-Condition	Patient is signed in. Patient has made at least one booking.
Basic Flow	
Patient Action	**System Response**

1: Navigates to the "My Lab bookings" Section. **3:** Clicks on one of the buttons.	**2:** Displays bookings made by patient and their status and action buttons. **4:** Initiates respective Use case.
Alternative Flow	
2A: If no booking found system will show empty booking page saying No booking found.	
Post-Condition	Patient views his bookings.
Extension	
Use case	**Condition**
Accept Booking	Triggered when a patient tries to click on the accept button.
Modify Booking	Triggered when a patient tries to click on the modify button.
Cancel Booking	Triggered when a patient tries to click on the cancel button.

4.4.2.8 Accept a Booking (Patient)

Table 4.16: Accept a booking FD

UC-ID	P-07-A
Use Case Name	Accept Booking
Primary Actor	Patient
Purpose	To allow the Patient to accept a booking .
Pre-Condition	1. MLO modified the booking at-least once.
Basic Flow	

MLO Action	System Response
1: Clicks on “Accept” button for a booking.	**2:** Updates the booking status to "Accepted," notifies the MLO, and prevents further modifications by lab or patient.
Alternative Flow	
1A: If Patient attempt to accept booking before Lab’s modification, system shows error “ cannot perform operation, please wait for lab response first”.	
Post-Condition	The Patient has accepted a booking.

4.4.2.9 Modify a Booking(Patient)

Table 4.18: Modify a booking FD (Patient)

UC-ID	P-07-B
Use Case Name	Modify Booking
Primary Actor	Patient.
Purpose	To allow the Patient to Modify a booking.
Pre-Condition	1. Patient has clicked on Modify button for any booking in manage lab booking flow.
Basic Flow	
Patient Action	**System Response**
1: Clicks on “Modify” button for a booking. **3:** Sets new date and time and proceeds.	**2:** Shows modification options(date and time) **4:** Saves the changes , update booking details.
Alternative Flow	

1A: If Patient tries to modify an accepted booking system will display “cannot modify accepted bookings”.	
Post-Condition	The Patient has Modified a booking.

4.4.2.10 Cancel Booking (Patient)

Table 4.18: Cancel Booking FD (Patient)

UC-ID	P-07-C	
Use Case Name	Cancel Booking.	
Primary Actor	Patient.	
Purpose	To allow patient cancel a lab booking.	
Pre-Condition	1. Patient must click on cancel button for any booking from the Manage Lab Bookings flow.	
Basic Flow		
Patient Action		**System Response**
1: Clicks on “Cancel” button for a booking.		**2:** Updates the booking status to "Canceled" and removes the booking from both ends.
Alternative Flow		
1A: If Patient tries to Cancel an accepted booking, system will show “cannot proceed, booking has already been accepted”.		
Post-Condition	The Patient has cancelled a booking.	

4.4.2.11 Payment

Table 4.19: Payment FD Uc

UC-ID	P10
Use Case Name	Payment
Primary Actor	Patient.
Purpose	To allow the patient to simulate payment for lab bookings, nurse bookings, or medicine orders using a dummy payment system (Stripe).

Pre-Condition	1. The patient must be signed in. 2. The patient must have at least one confirmed booking or order.
Basic Flow	
Patient Action	**System Response**
1. Confirms a lab booking, nurse booking, or medicine order. **3.** Clicks on the “Pay Now” button. **5.** Enters dummy payment details (e.g., test card information) and clicks on the “Pay” button.	**2.** Displays the corresponding bill, including itemized costs (e.g., service fees, taxes). **4.** Redirects the patient to a Stripe dummy payment screen. **6.** Upon successful simulation, the system confirms the transaction and sends a notification to both the patient and the service provider.
Alternative Flow	
5A: If the patient exits the dummy payment screen without completing the simulation, the system cancels the simulated transaction and notifies the patient: "Payment not completed. Please confirm your order or booking again to retry."	
Post-Condition	The patient successfully simulates payment and the service provider is notified.

4.4.2.12 Patient Manage Nurse Booking.

Table 4.20:Manage Nurse Booking

UC-ID	P11
Use Case Name	Manage Nurse Booking.

Primary Actor	Patient.
Purpose	To allow the patient Accept/reject bid from a nurse.
Pre-Condition	1. Patient is signed in. 2. Patient has an existing nurse booking.
Basic Flow	
Patient Action	**System Response**
1. Clicks on any nurse booking. **3.** Accepts any nurse bid.	**2.** Displays Booking details including bids. **4.** Assign the booking to that nurse and notify the nurse about acceptance.
Alternative Flow	
2A: If there isn't any bid present, system will ask user to wait for nurse to bid. **5A:** If patient rejects any bid, the system will notify the nurse and will allow the nurse to bid again.	
Post-Condition	Patient has accepted a nurse's bid for booking successfully.

4.4.2.13 View Test (Lab)

Table 4.21: Manage Test FD

UC-ID	ML01
Use Case Name	Manage Test
Primary Actor	Medical Laboratory Owner
Purpose	To allow MLO to manage test.
Pre-Condition	1. MLO must be logged in.
Basic Flow	
MLO Action	**System Response**
1: Selects the "View Tests" option. **3:** Makes his choice.	**2:** Displays a list of tests offered by the lab with an "Add Test" button and delete button and edit button with all tests. **4:** Initiate respective Use case.

Post-Condition	MLO has managed his tests successfully.
Use case	**Condition**
Add a test	MLO selects add test option.
Delete a test	MLO selects delete option.

4.4.2.14 Add a Test (Lab)

Table 4.22: Add lab test FD Uc

UC-ID	ML01-A
Use Case Name	Add a test
Primary Actor	Medical Laboratory Owner
Purpose	To allow MLO to add a test.
Pre-Condition	1. MLO must be logged in.
Basic Flow	
MLO Action	**System Response**
1: Selects the "view test list" option. **3:** Selects a test. **4:** Sets price and click Add Service.	**2:** Displays a dropdown of tests and ask MLO to select. **4:** System ask MLO to set price. **5:** Adds the test to labs test list and display confirmation.(service added successfully).
Alternative Flow	
4A: If MLO tries to add a single test for the second time, system will display error message " couldn't add test as test already added".	
Post-Condition	MLO has added a test listing.

4.4.2.15 Delete a Test (Lab)

Table 4.23: Delete Test FD Uc

UC-ID	ML01-B
Use Case Name	Delete Test.

Primary Actor	Medical Laboratory Owner
Purpose	To allow MLO to delete a test.
Pre-Condition	1. Test must be there to delete.
Basic Flow	
MLO Action	**System Response**
1: Clicks on delete icon from any test listing. **3:** Confirms the deletion.	**2:** Ask for confirmation. **4:** Deletes the selected test.
Alternative Flow	
3A: If the MLO cancels the deletion, the system goes back to test list.	
Post-Condition	MLO has deleted a test listing.

4.4.2.16 Manage Booking (Lab)

Table 4.24: Manage Bookings FD Uc

UC-ID	ML02
Use Case Name	Manage Bookings
Primary Actor	Medical Laboratory Owner
Purpose	To allow MLO to manage bookings and their confirmations.
Pre-Condition	1. MLO must be verified 2. MLO must have at least one booking.
Basic Flow	
MLO Action	**System Response**
1: Navigates to "view booking" section. **3:** Chooses any option from the buttons:	**2:** System shows bookings made by patients with accept, modify and reject buttons with each booking. **4:** System initialize the respective use case.

Alternative Flow	
2A: If any record not found against bookings then system shows “ No Booking Found”.	
Post-Condition	MLO has managed his bookings.
Extensions	
Use case	**Condition**
Accept booking	If patient click on accept button from a booking.
Modify Booking	If patient click on modify button from a booking.
Cancel Booking	If patient click on cancel button from a booking.

4.4.2.17 Accept Booking

Table 4.25: Accept Booking FD

UC-ID	ML-02-A
Use Case Name	Accept Booking
Primary Actor	MLO
Purpose	To allow the MLO to accept a booking.
Pre-Condition	1. MLO has clicked on accept button from any booking.
Basic Flow	
MLO Action	**System Response**
1: Clicks on “Accept” button for a booking.	**2:** Updates the booking status to "Accepted," notifies the Patient, and prevents further modifications by lab or patient.
Alternative Flow	
1A: If lab has modified booking and tries to accept, system display “cannot perform operation now, wait for patient’s response”.	
Post-Condition	The MLO has accepted a booking.

4.4.2.18 Modify Booking

Table 4.26: Modify a booking FD

UC-ID	ML-02-B	
Use Case Name	Modify Booking	
Primary Actor	MLO	
Purpose	To allow MLO to Modify a booking.	
Pre-Condition	1. Patient has clicked on Modify button for any booking in manage lab booking flow.	
Basic Flow		
MLO Action		**System Response**
1: Clicks on "Modify" button for a booking. **3:** Sets new date and time and proceeds.		**2:** Shows modification options(date and time) **4:** Saves the changes , update and notify patient.
Alternative Flow		
1A: If MLO tries to modify an accepted booking system will display "cannot modify accepted bookings".		
Post-Condition	The MLO has Modified a booking.	

4.4.2.19 Reject a Booking

Table 4.27: Reject a booking FD (MLO)

UC-ID	ML-02-C
Use Case Name	Reject Booking
Primary Actor	MLO
Purpose	MLO has clicked on reject button from any booking.
Pre-Condition	2. MLO must click on Reject button for any booking in the Manage booking flow .
Basic Flow	

MLO Action	System Response
1: Clicks on “Reject” button for a booking.	**2:** Updates the booking status to "Rejected," notifies the Patient, and removes the booking from both ends.
Alternative Flow	
1A: If MLO tries to Reject an accepted booking, system will show “cannot proceed, booking has already been accepted”.	
Post-Condition	The MLO has Rejected a booking.

Nurse

4.4.2.20 Nurse View Booking

Table 4.28: Nurse Manage Booking FD Uc

UC-ID	N01
Use Case Name	Manage Booking.
Primary Actor	Nurse
Purpose	To allow Nurse to manage booking requests(bid for them).
Pre-Condition	1. Nurse must be logged in. 2. Nurse must be verified.
Basic Flow	
Nurse Action	**System Response**
1: Navigates to View Booking request page.	**2:** Shows nearby request of patients with service request info.
3: Clicks on place a bid for any request.	**4:** Shows the bid slider and ask nurse to set a bid.
5: Set price and click on proceed.	**6:** Saves the bid and shows it to patient and saves update.

Alternative Flow	
3A: If the Nurse leaves the page without saving system will discard all changes.	
Post-Condition	Nurse made changes to his profile.

4.4.2.21 Nurse View Bookings

Table 4.29: Nurse View Bookings FD uc

UC-ID	N02
Use Case Name	View Booking history
Primary Actor	Nurse
Purpose	To allow Nurse view his bookings.
Pre-Condition	1. Nurse must be logged in. 2. Nurse must be verified.
Basic Flow	
Nurse Action	**System Response**
1: Navigates to view bookings page. **3: V**iews its past booking.	**2:** Shows nurse's past bookings list. **4:** Shows booking details.
Alternative Flow	
2A: If no bookings available system will show list empty.	
Post-Condition	Nurse views his bookings history.

4.4.2.22 Nurse Complete Bookings

Table 4.30: Nurse View Bookings FD uc

UC-ID	N02
Use Case Name	Mark booking as Complete
Primary Actor	Nurse

Purpose	To allow Nurse complete his booking.
Pre-Condition	3. Nurse must be logged in. 4. Nurse must be verified.
Basic Flow	
Nurse Action	**System Response**
1: Navigates to view bookings page. **3:** Select any booking request. **5:** Mark booking as Complete. **7:** Select payment method.	**2:** Shows nurse's bookings list. **4:** Shows booking details. **6:** Shows payment option **8:** Shows payment details according to select ed payment method.
Alternative Flow	
2A: If no bookings available system will show list empty.	
Post-Condition	Nurse mark the booking completed.

Medical Store

4.4.2.23 Store Respond to Order Request

Table 4.31: Store Respond to Order FD Uc

UC-ID	MS-01
Use Case Name	Respond to Order Request.
Primary Actor	Medical Store owner
Purpose	To allow the medical store owner to accept or ignore a medicine delivery order.
Pre-Condition	1. The medical store owner must be logged into the system. 2. The store must be verified.
Basic Flow	

Store Action	System Response
1: Navigates to order requests section. **3:** Selects a request. **5:** Make price bid for patient. **7:** accept any medical store bid that suits.	**2:** Display patient's medicine delivery requests. **4:** Shows request details(medicine list , request type, and prescription(if uploaded)). **6:** Notifies patient about the stores bid and ask the user to perform action. **8:** Display order confirmation message on screen the store will prepare your order and contact you soon.
Alternative Flow	
1A: If the store is not verified, the system prevents access to the "Orders" section and displays: "Your store must be verified to manage orders. Please verify your account." **3A:** If any other medical store accepts the order system will remove the request from every other store's requests section.	
Post-Condition	The order is accepted or ignored by a particular store.

4.4.2.24 Store View Order Request

Table 4.32: Store View Order

UC-ID	MS-02
Use Case Name	View Order Requests.
Primary Actor	Medical Store owner

Purpose	To allow the medical store owner to view medicine orders and their respective prescriptions.
Pre-Condition	1. The store must be verified.
Basic Flow	
Store Action	**System Response**
1: Navigates to view order requests section. **3:** Selects a request. **5:** respond to request of selected request.	**2:** Display patient's medicine delivery requests. **4:** Shows request details(medicine list , request type, and prescription(if uploaded)). **6:** send bid notification to patient.
Alternative Flow	
1A: If the store is not verified, the system prevents access to the "Orders" section and displays: "Your store must be verified to manage orders. Please verify your account."	
Post-Condition	The medical store owner successfully views order and the attached prescription for the selected order.

4.4.2.25 Store Modify Order Status

Table 4.33: Store Modify Order Status FD Uc

UC-ID	MS-03

Use Case Name	Modify Order Status
Primary Actor	Medical Store owner
Purpose	To allow the medical store owner to update the status of an order (e.g., processing, delivered, not delivered).
Pre-Condition	1. The order must exist in the system and belong to the store.

Basic Flow	
Store Action	**System Response**
1: Navigates to the order management section.	**2:** Displays a list of orders associated with the store.
3: Selects an order and clicks on "Modify Status."	**4:** Shows the current status of the order and available options (e.g., processing, Delivered, not delivered).
5: Selects a new status and confirms the change.	**6:** Updates the order status in the database.
	7: If the status is changed to "Delivered," the system saves the time taken and compare it with decided time and label the order.
	9: If medical store set the status to "not delivered" system will label the order as incomplete and ask store for reason.

Alternative Flow
5: If the medical store owner does not confirm the new status, the system retains the previous status and notifies the owner that the status was not updated.

Post-Condition	The order status is updated successfully.

Admin

4.4.2.26 Admin Login

Table 4.34: Admin Login FD Uc

UC-ID	AD-01
Use Case Name	Login
Primary Actor	Admin
Purpose	To allow the Admin to log into the system.
Pre-Condition	3. The admin must have correct credentials.
Basic Flow	
Admin Action	**System Response**
1: Navigates to the login page. **3:** Enters the correct username and password. **5:** Clicks on the "Login" button.	**2:** Displays the login form. **4:** Validates the credentials. **6:** Grants access to the Admin dashboard.
Alternative Flow	
3A: If the Admin enters invalid credentials, the system displays an error message: "Invalid username or password. Please try again.	
Post-Condition	The Admin is logged into the system and redirected to the dashboard.

4.4.2.27 Admin Manage providers

Table 4.35: Admin Manage Providers FD Uc

UC-ID	AD-03
Use Case Name	Manage Service Providers.
Primary Actor	Admin
Purpose	To allow the Admin to monitor service providers' performance, including feedback and blacklist/whitelist them.
Pre-Condition	1. The Admin must be logged into the system.
Basic Flow	
Admin Action	**System Response**
1: Navigates to the "Manage Service Providers" section. **3:** Selects a service provider to review their details. **5:** Clicks "Blacklist" or "Whitelist" for the provider. **7:** Confirms the action.	**2:** Displays a list of service providers. **4:** Shows performance information and status (e.g., Active, Blacklisted) and their performance metrics (e.g., feedback ratings, **6:** Asks for confirmation: "Are you sure you want to [Blacklist/Whitelist] this provider?" **8.** Updates the provider's status accordingly.
Alternative Flow	
3A: If performance data is unavailable, the system prompts: "Performance data is not available at the moment."	
Post-Condition	The service provider's status is updated (Blacklisted or Whitelisted), and it is reflected in the system.

4.4.2.28 Admin Manage Profile

Table 4.36: Admin Manage Profile FD Uc

UC-ID	AD-04	
Use Case Name	Manage Profile	
Primary Actor	Admin	
Purpose	To allow the Admin to update their profile information.	
Pre-Condition	1. The admin must have correct credentials.	
Basic Flow		
Admin Action		**System Response**
1: Navigates to the "Profile" section. **3:** Clicks on "Edit Profile." **5:** Clicks on the "Login" button.		**2:** Displays the Admin's current profile details. **4:** Shows editable fields (e.g., name, email, password).. **6:** Updates profile details and saves.
Alternative Flow		
5A: If invalid input is entered, the system prompts: "Invalid details. Please correct and try again."		
Post-Condition	The Admin's profile is successfully updated in the system.	

4.4.2.29 Admin Logout

Table 4.37: Admin Logout FD Uc

UC-ID	AD-05
Use Case Name	Logout.
Primary Actor	Admin
Purpose	To allow the Admin to logout from the system.
Pre-Condition	1. The admin must be logged in.

Basic Flow	
Admin Action	**System Response**
1: Navigates to his profile. **3:** Clicks on logout. **5:** Clicks on confirm.	**2:** Displays the profile. **4:** Asks for confirmation. **6:** Logs admin out of cura link.
Alternative Flow	
4A: If the Admin decides not to confirm the logout: The system cancels the logout operation and retains the Admin session. The Admin is redirected back to the profile page.	
Post-Condition	The Admin is logged out from his panel and redirected to login page.

4.5 Activity Diagrams

Below we have activity diagrams for Cura-Link.

4.5.1 Registration

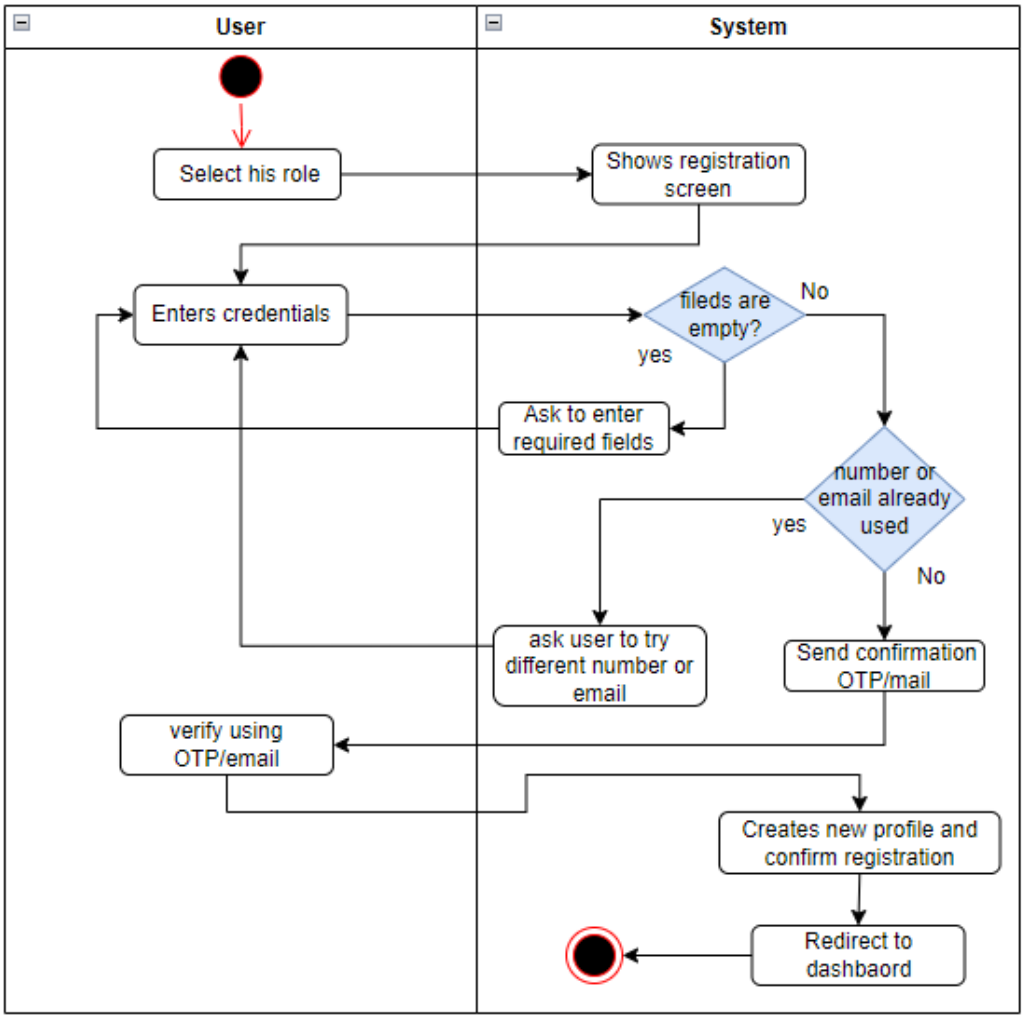

Figure 4.8: Registration Activity Diagram

4.5.2 Email Verification

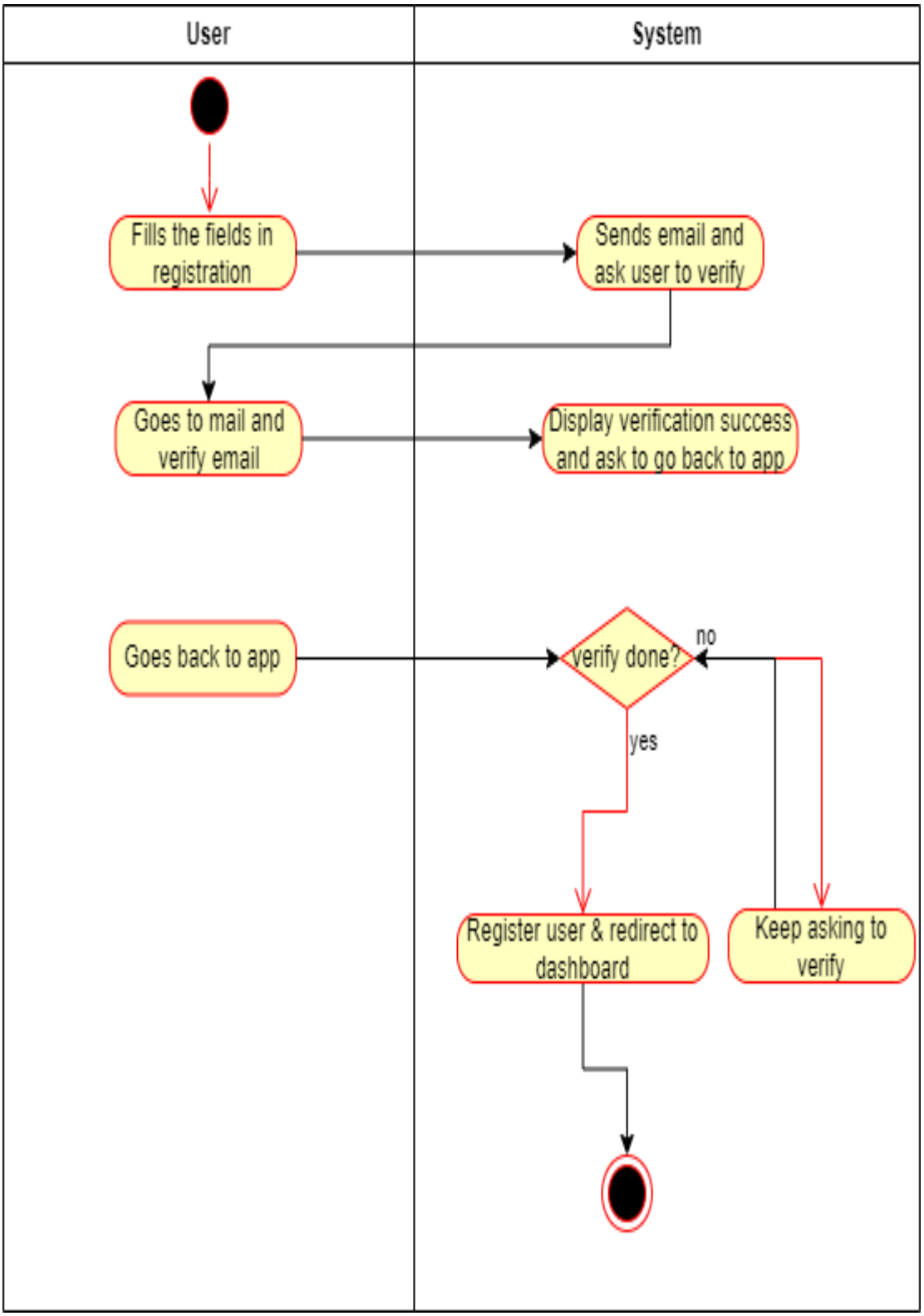

Figure 4.9: Email Verification Activity

4.5.3 Login

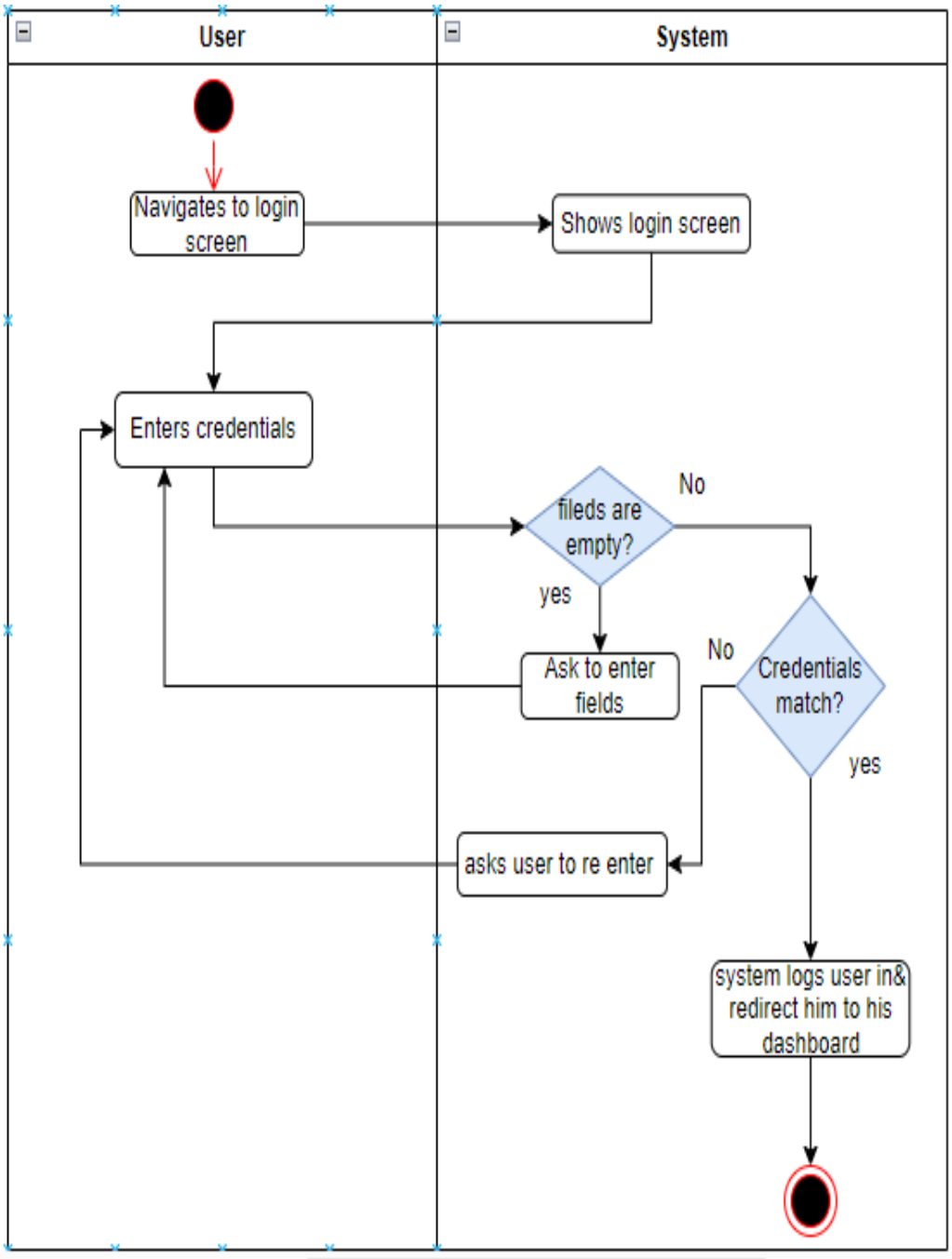

Figure 4.10: Login Activity Diagram

4.5.4 Forget Password

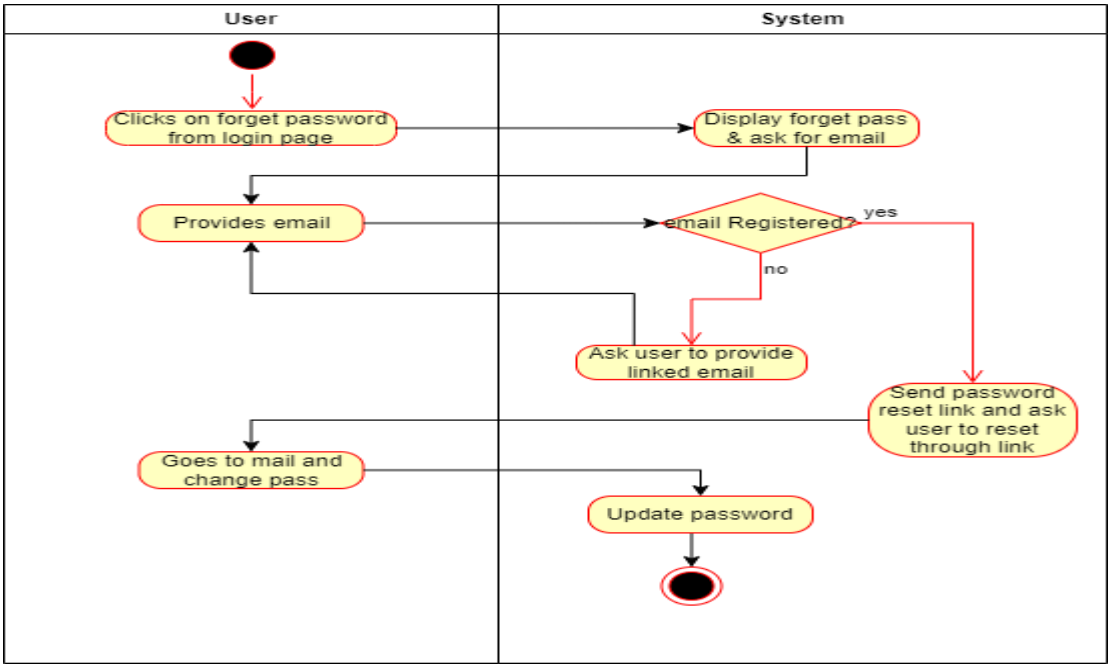

Figure 4.11: Forget Password Activity

4.5.5 Logout

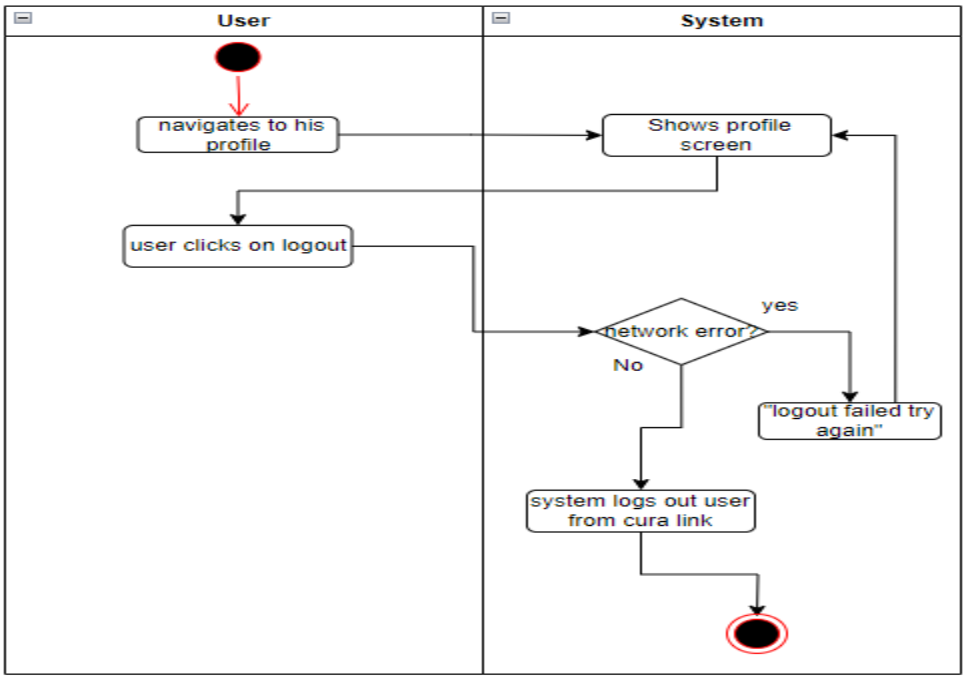

Figure 4.12: Logout Activity

4.5.6 Chat

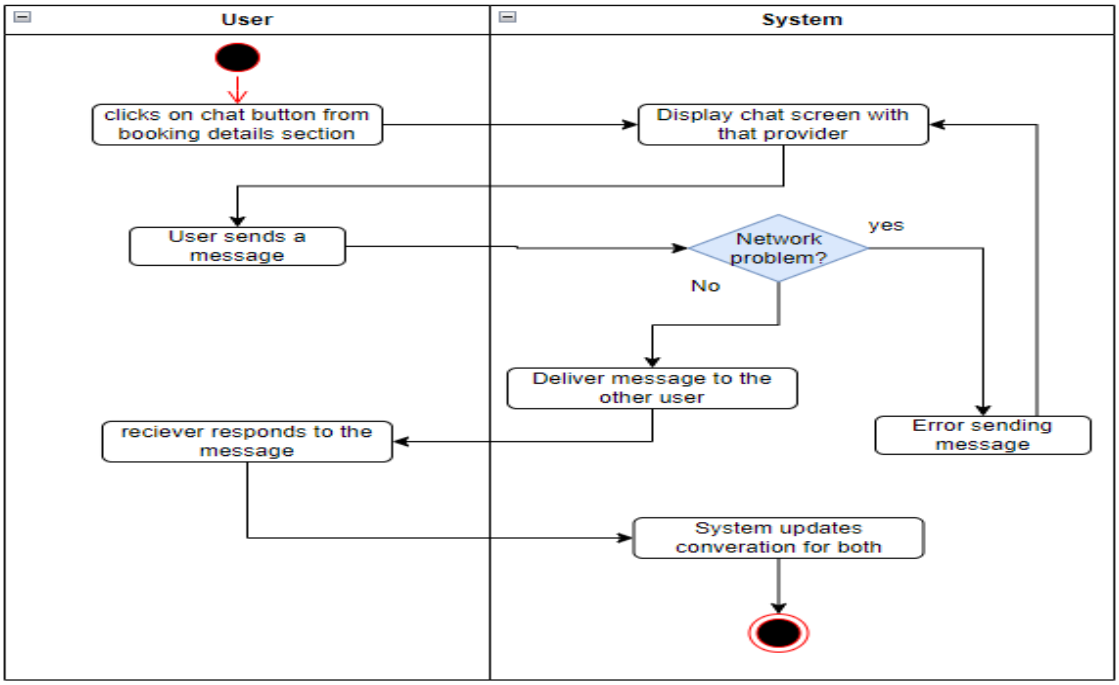

Figure 4.13: Chat Activity

4.5.7 Verification

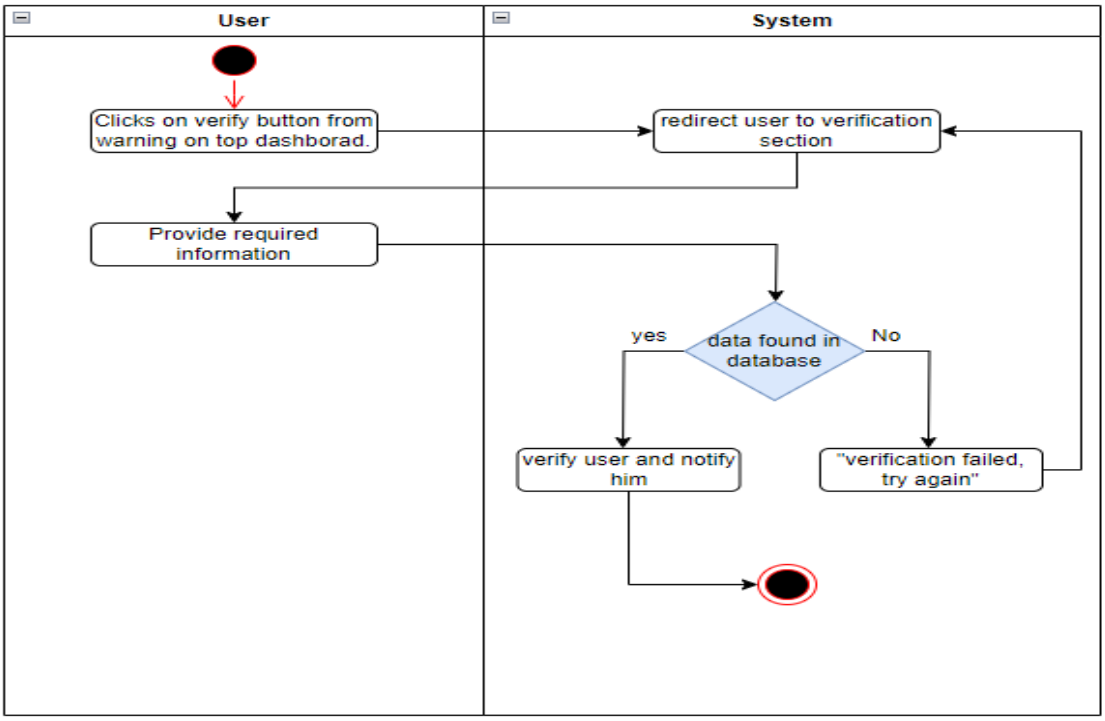

Figure 4.14: Verification Activity

4.5.8 Edit Profile

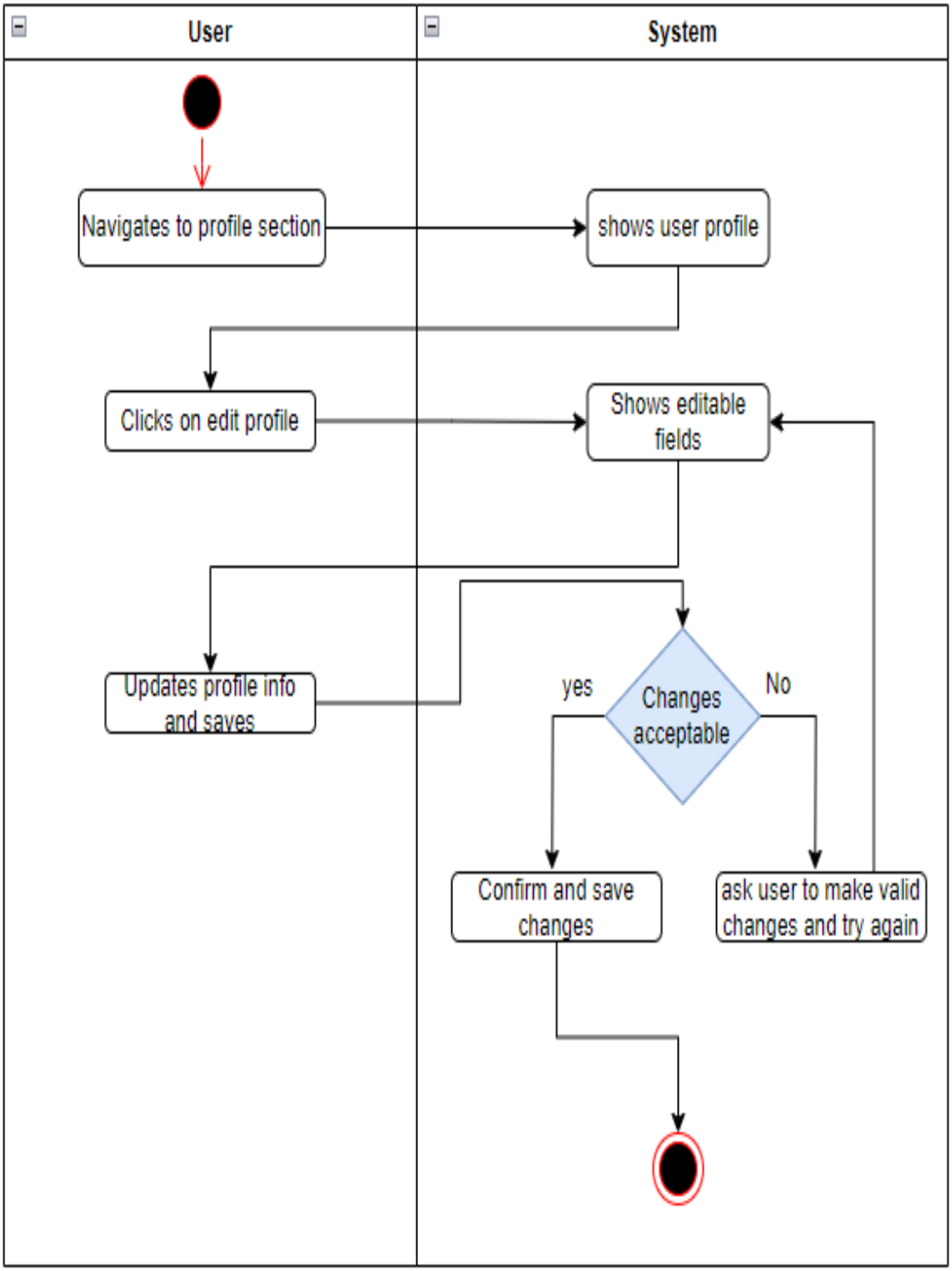

Figure 4.15: Edit Profile Activity

4.5.9 Respond to Feedback

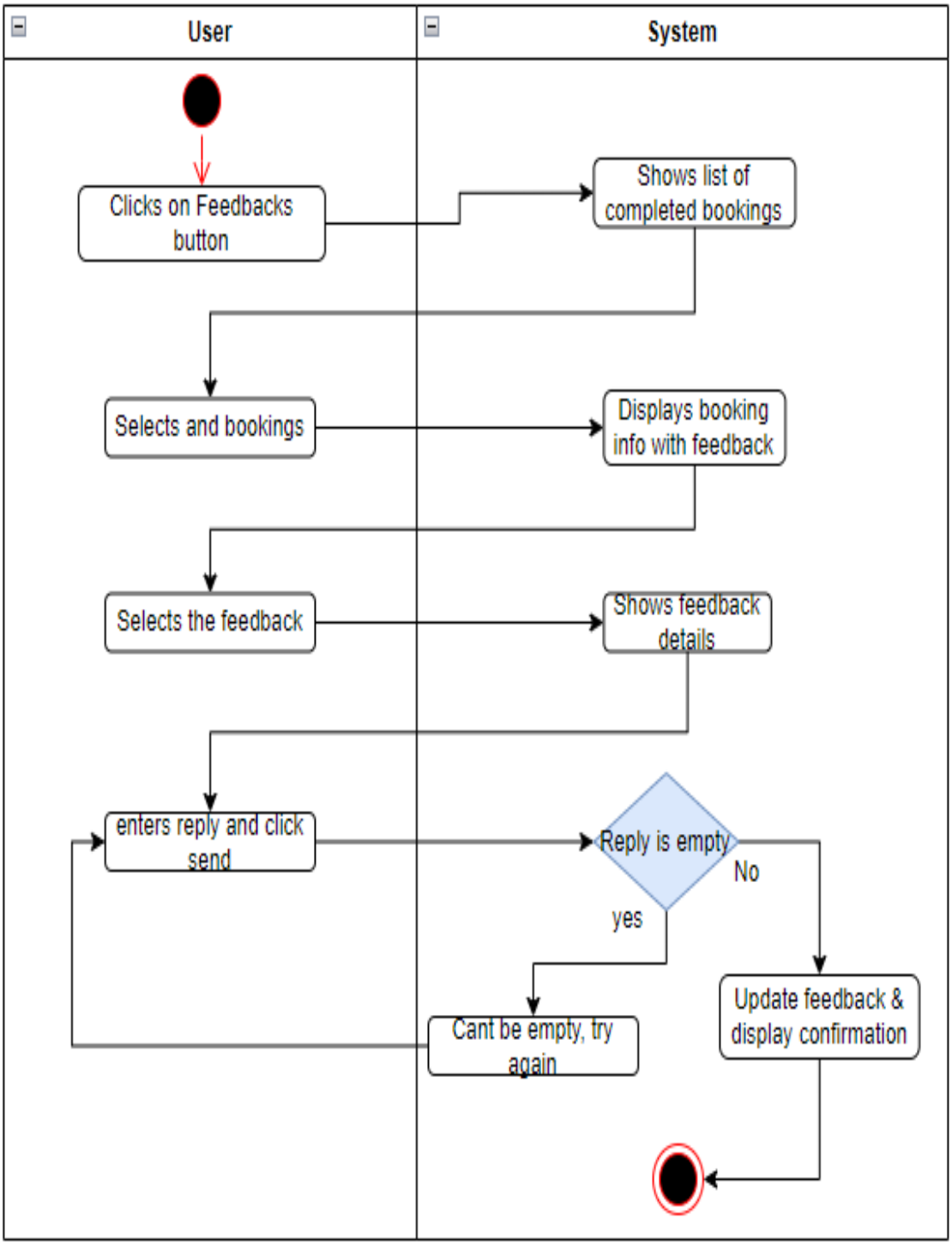

Figure 4.16: Respond to Feedback Activity

4.5.10 Order Medicine

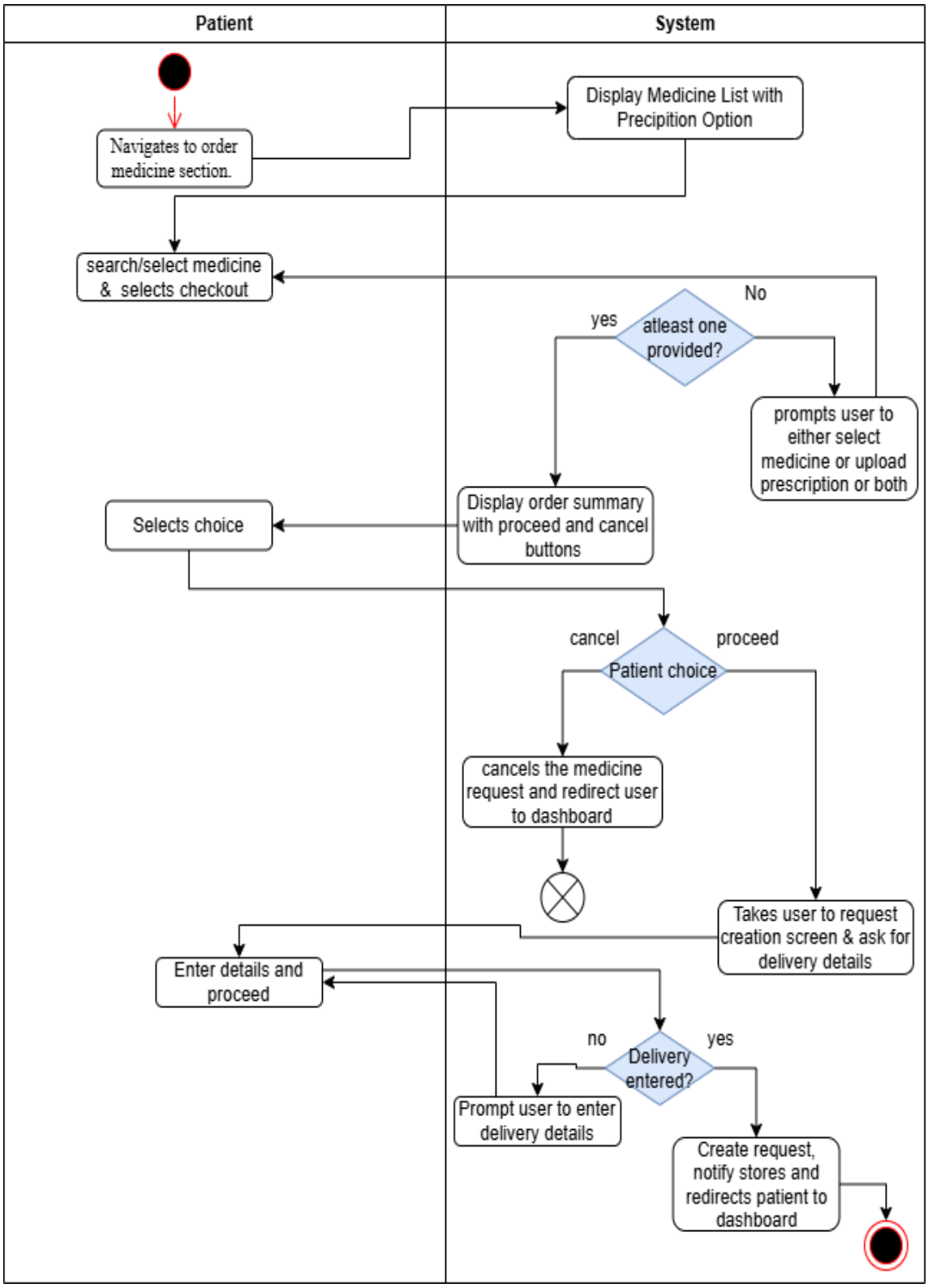

Figure 4.17: Order medicine Activity

4.5.11 Nurse Booking

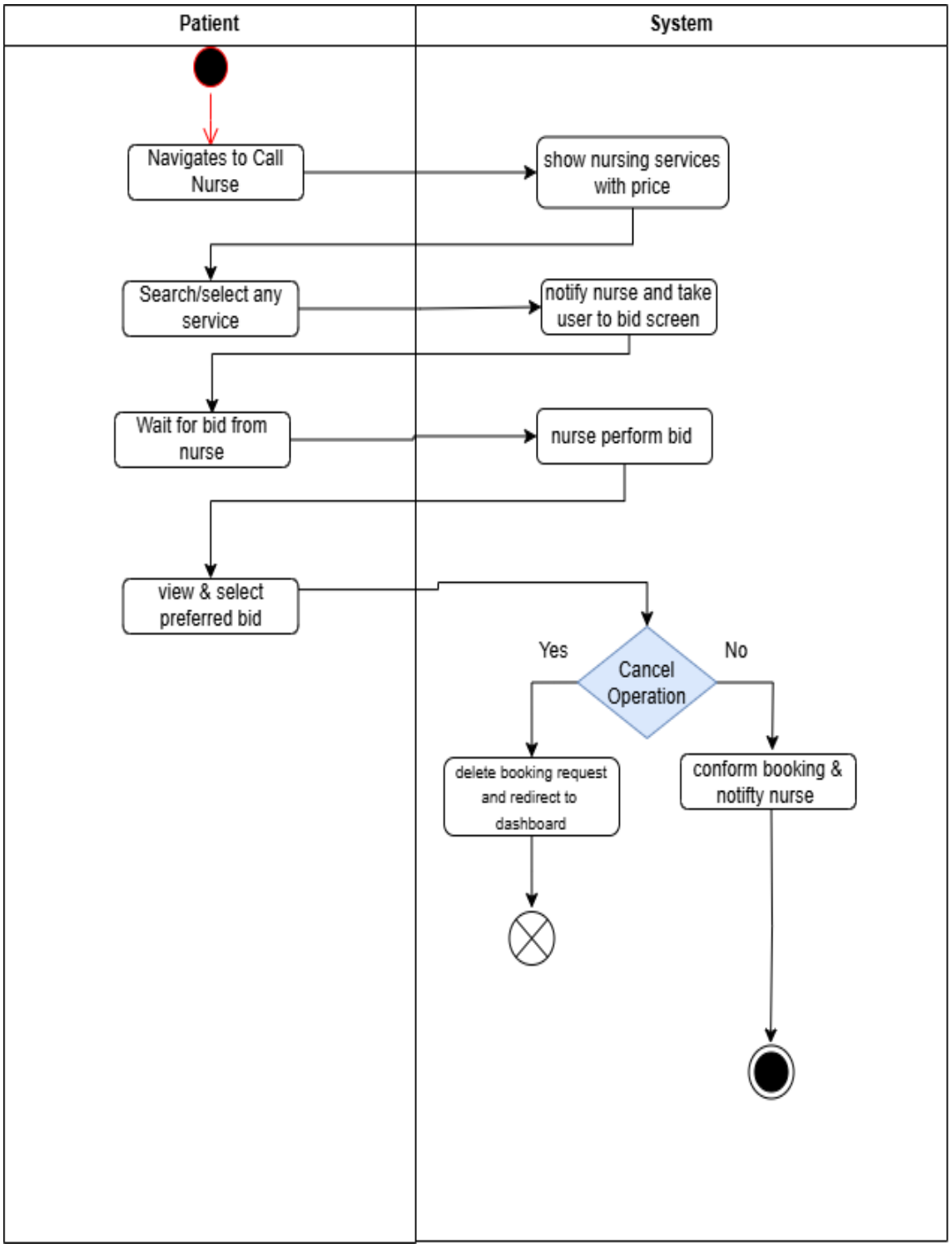

Figure 4.18: Bid Activity

4.5.12 Upload a Prescription

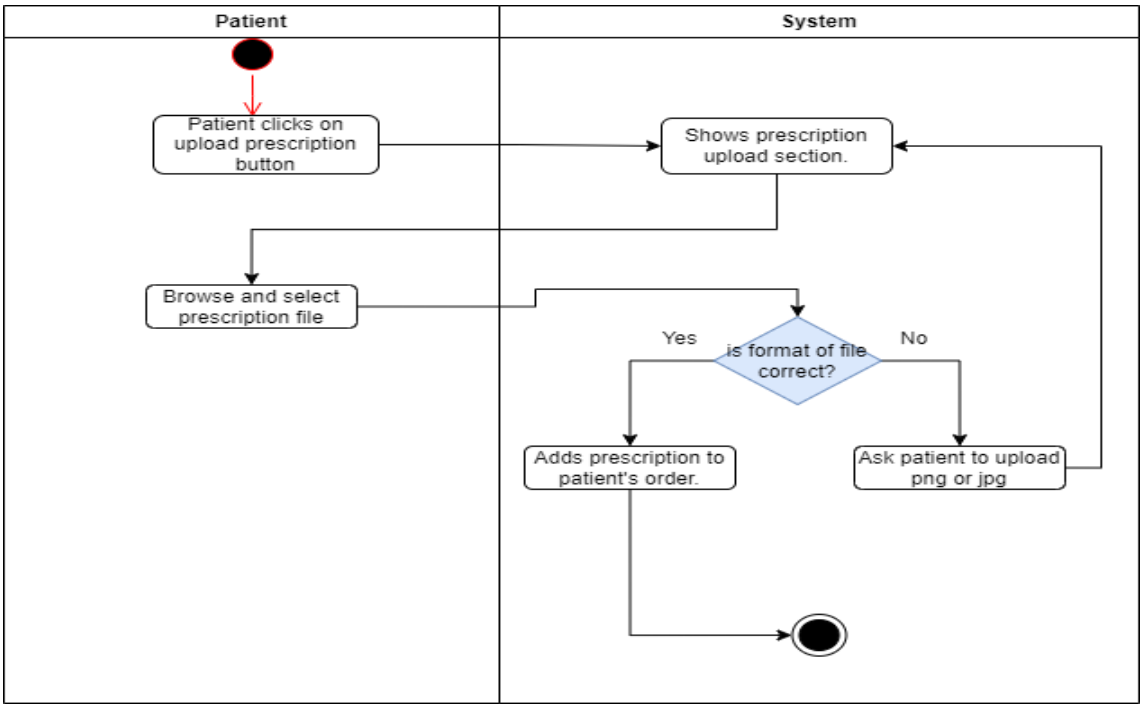

Figure 4.19: Prescription Upload Activity

4.5.13 Rate & Feedback

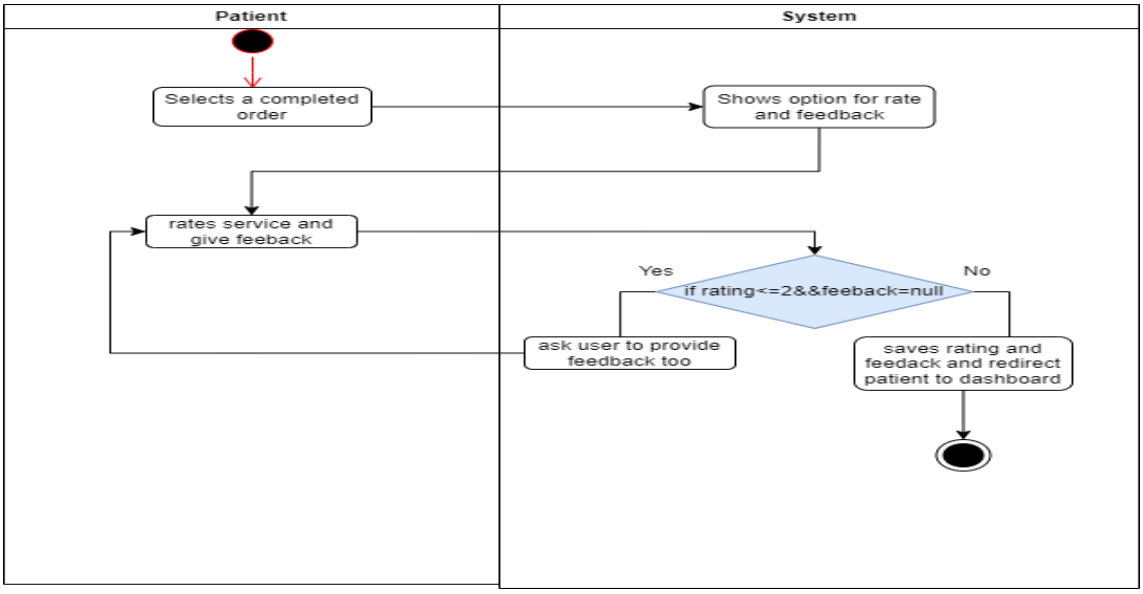

Figure 4.20: Rate & Feedback Activity

4.5.14 Book lab

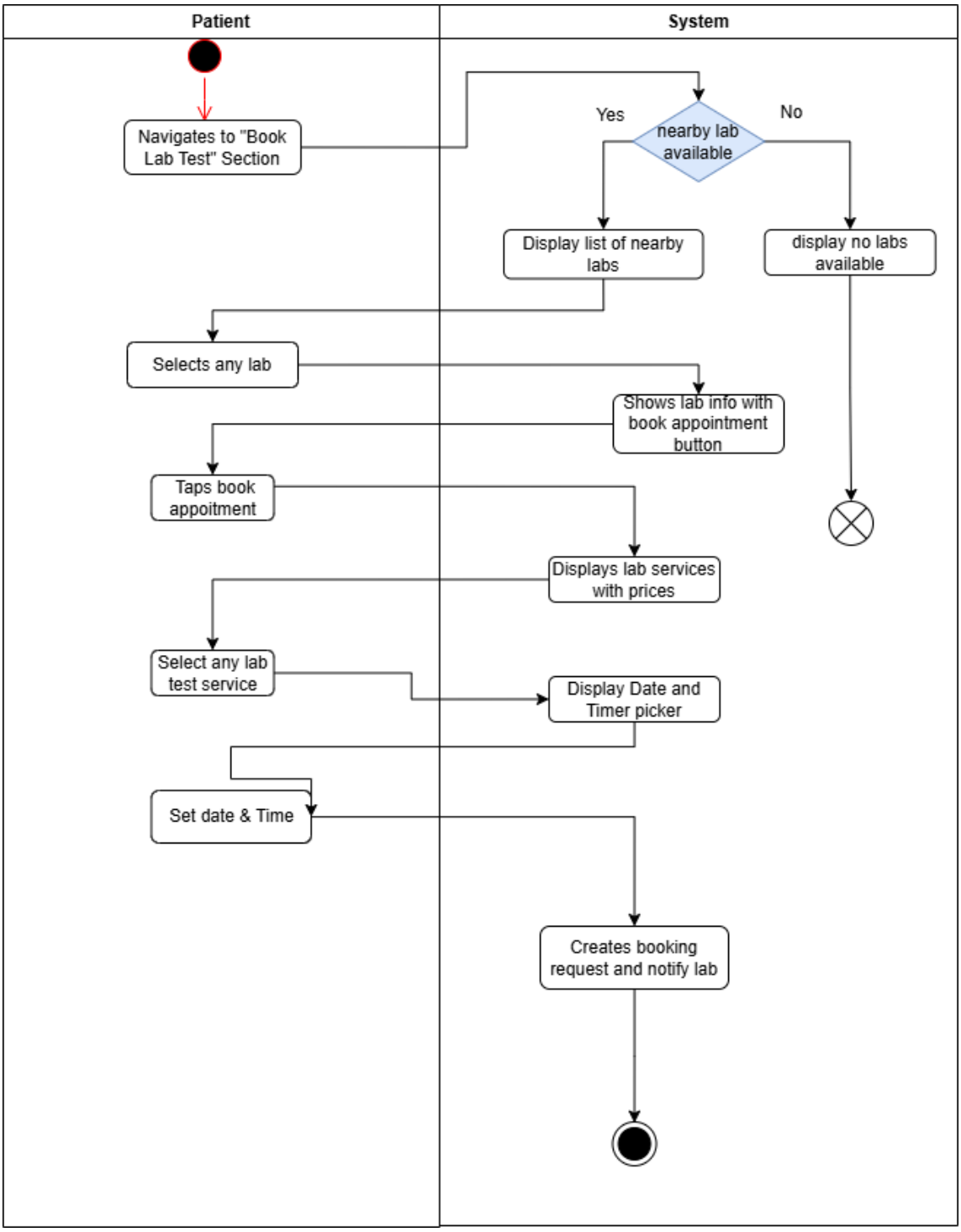

Figure 4.21: Book Lab Test Activity

4.5.15 View Orders (Patient)

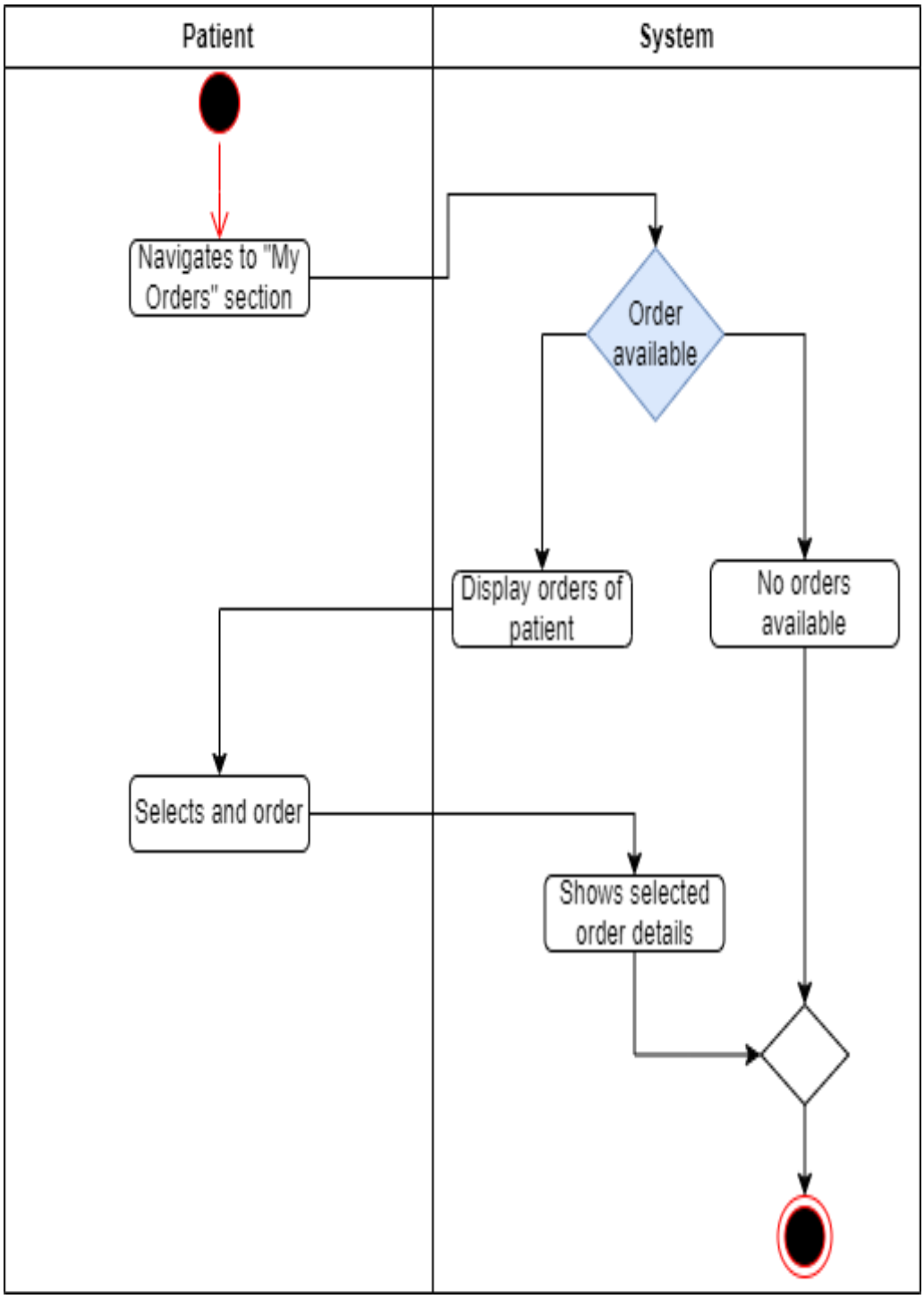

Figure 4.22: Patient View Order Activity

4.5.16 View Lab Bookings (Patient)

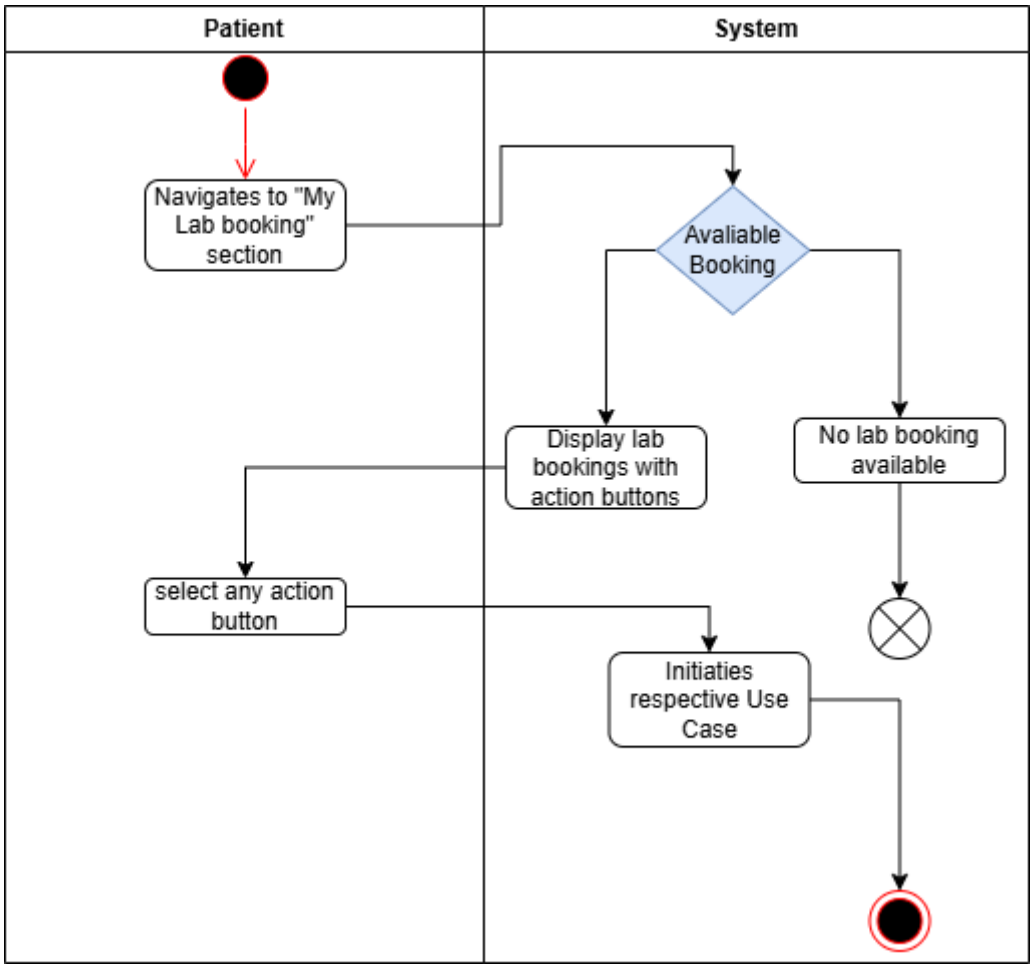

Figure 4.23: Manage Lab Booking(Patient) Activity

4.5.17 Payment

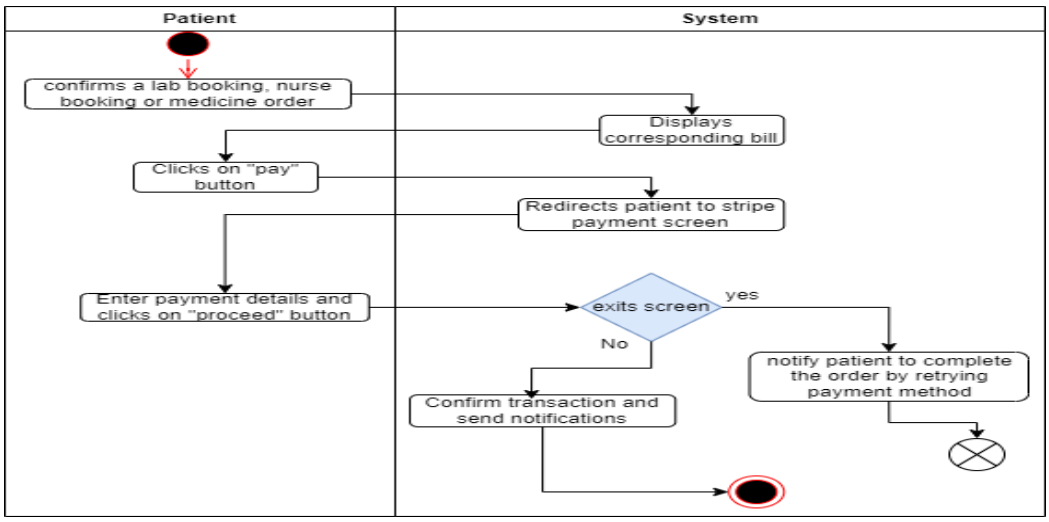

Figure 4.24: Payment Activity

4.5.18 Respond to Nurse Bids

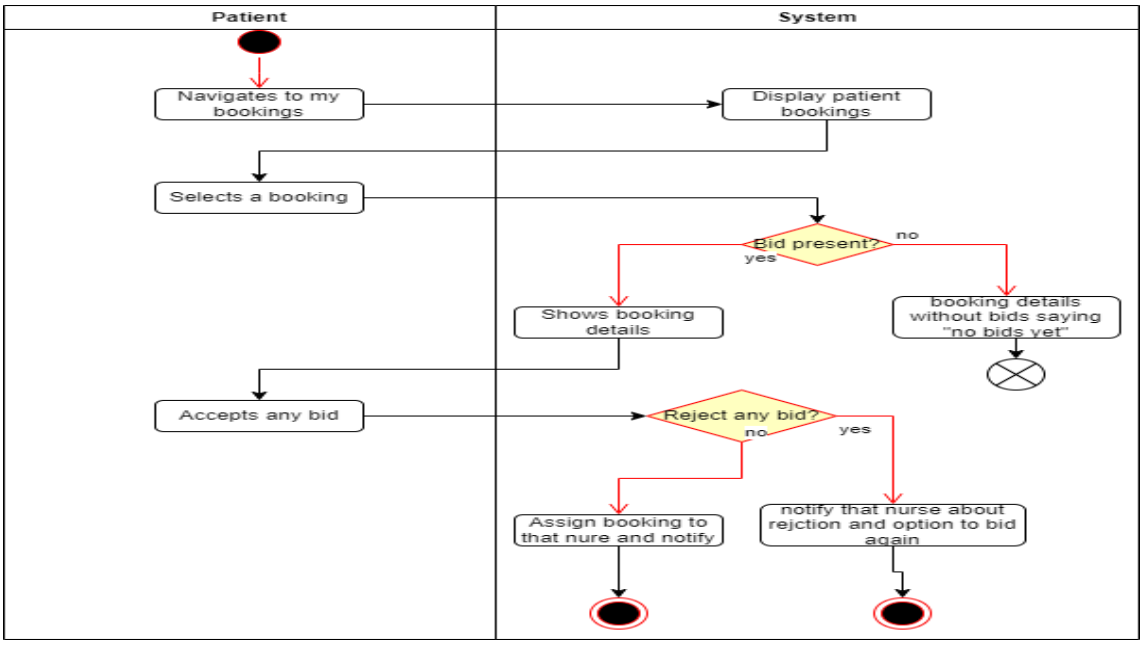

Figure 4.25: Respond to Bids Activity

4.6 Test Case Design(Black Box)

Below we have black box test cases.

4.6.1 User Registration

4.6.1.1 ECP Classes

Input (Functionality): Name, Email, Phone-no, Password.

- **Valid Input Class:**

  Full-name (Aa,….Zz), **Email** (Aa,…Zz)(0,1,…9)(@),

  Phone-no (0,1,…,9)

- **Invalid Input Class:**

  Full-name (input except valid class)**, Email** (input except valid class),

  Phone-no (input except valid class)

Test Case:

Table 4.38: User registration test case

Test ID	Test Title	Input (Email)	Phone-no	Full name	ECP Class	Expected Output
TC_009	Valid Sign up through Email	mnkj315@gmail.com	+923155875170	Nouman khan	Valid Input Class	Sigup Successfully
TC_010	Invalid email signup	mnkj315##.com	+923155875170	Nouman khan	Invalid input class	Error: Invalid email

4.6.2 User Login

4.6.2.1 ECP Classes

Input (Functionality): Email, Password.

- **Valid Input Class:**

  Email (Aa,…Za)(0,1,…9)(@)

- **Invalid Input Class:**

  Email (input except valid class)

Test Case:

Table 4.39: User Login Test case

Test ID	Test Title	Input (Email)	ECP Class	Expected Output
TC_009	Login with valid email	Mnkj315@gmail.com	Valid Input Class	Login Successfully
TC_010	Invalid email login attempt	Mnkj315##.com	Invalid Class	Invalid email format
TC_011	Email not registered	Usama123@gmail.com	Invalid Class	No Accont Exist

4.6.3 Book A Lab Test

4.6.3.1 ECP Classes

Input (Functionality): Test-type, Date, Time.

- **Valid Input Class:**

  Test-type (selected), **Date** (selected), **Time** (Selected)

- **Invalid Input Class:**

  Test-type (not selected), **Date** (not selected), **Time** (not selected)

Test Case:

Table 4.40: Book a lab test case

Test ID	Test Title	Input (test-type)	date	Time	ECP Class	Expected Output

TC_009	Book test valid	selected	selected	selected	Valid Input Class	Bookig Successfully
TC_010	Invalid book test	selected	Not seleted	selected	Invalid input class	Please select date
TC_011	Invalid book test		selected	selected	Invalid input class	Please select test type
TC_012	Invalid book test	selected	selected		Valid input class	Please select \Time

4.6.4 Upload Prescription

4.6.4.1 ECP Classes

Input (Functionality): File Name, File Size, Username, Password

- **Valid Input Class:**

  FileName: {a-z, A-Z, 0-9, ., pdf, jpg, png},**File Size:** {1MB, 2MB, 3MB, ...},**Username:** {a-z, A-Z, 0-9, _},Password: {a-z, A-Z, 0-9, _, @, $, %, !}

- **Invalid Input Class:**

  File Name: {*, &, %, $, #, @},**File Size:** {0MB, -1MB, -2MB, ...},**Username:** {#, @, $, %},**Password**: {*, &, %, $, #, @}

Test Case:

Table 4.41: Upload Prescription Test case

Test ID	Test Title	Input	ECP Class	Expected Output

TC_009	Upload Prescription	prescription123.png, 2MB curalink, SecurePass123	Valid Input Class	Prescription successfully uploaded
TC_010	Invalid File Name	*prescription123.pdf 2MB Curalink SecurePass123	Invalid File Name Class	Error message for invalid file name
TC_011	Invalid File Size	prescription123.pdf, 0MB Curalink SecurePass123	Invalid File Size Class	Error message for invalid file size
TC_012	Invalid Username	prescription123.pdf, 2MB #curalink SecurePass123	Invalid Username Class	Error message for invalid username

4.6.5 Rating And Feedback

4.6.5.1 ECP Classes

Input (Functionality): Rating, Feedback, Username, Password

- **Valid Input Class:**

  Rating: {1, 2, 3, 4, 5},**Feedback:** {a-z, A-Z, 0-9, space, ., !, ?},**Username**: {a-z, A-Z, 0-9, _},**Password**: {a-z, A-Z, 0-9, _, @, $, %, !}

- **Invalid Input Class:**

Rating: {0, -1, 6, 7, ...},**Feedback:** {*, #, %, $, @},**Username:** {#, @, $, %}

Password: {*, &, %, $, #, @}

Test Case:

Table 4.42: Rating and Feedback Test case

Test ID	Test Title	Input	ECP Class	Expected Output
TC_009	Submit Rating and Feedback	6 Good service!, curalink SecurePass123	Valid Input Class	Rating and feedback successfully submitted
TC_010	Invalid Rating	5 Good service!, Curalink SecurePass123	Invalid Rating Class	Error message for invalid feedback
TC_011	Invalid Feedback	6 *Great service!, curalink SecurePass123	Invalid Feedback Class	Error message for invalid feedback
TC_012	Invalid Username	3 Good service!, #curalink SecurePass123	Invalid Username Class	Error message for invalid username
TC_017	Invalid Password	8 Good service!, curalink	invalid Password Class	Error message for invalid password

		@securePass123		

4.6.6 Medicine Order with Perception Upload

4.6.6.1 ECP Classes

Input (Functionality): Medicine ID, Quantity, Delivery Address, Order ID, File Name, File Size

- **Valid Input Class:**

  Medicine ID: {a-z, A-Z, 0-9},**Quantity:** {1, 2, 3, ...},**Delivery Address**: {A-Z, a-z, 0-9, space, #, @, ., ,},**Order ID**: {a-z, A-Z, 0-9, -},**File Name:** {a-z, A-Z, 0-9, ., pdf, jpg, png},**File Size:** {1MB, 2MB, 3MB, ...}

- **Invalid Input Class: Medicine ID**:

  {*, &, %, $, #, @},**Quantity:** {0, -1, -2, ...},**Delivery Address**: {*, %, $, !, &, #},**Order ID:** {*, @, #, $, %},**File Name**: {*, &, %, $, #, @},**File Size**: {0MB, -1MB, -2MB, ...}

Test Case:

Table 4.43: Prescription test case

Test ID	Test Title	Input	ECP Class	Expected Output
TC_009	Medicine Order with Valid Prescription	med123, 2 789pindiSt,Lahore ord456 prescription.png, 2MB	Valid Input Class	Medicine order placed with valid prescription

TC_010	Invalid Medicine ID with Prescription	#*med123, 2 789pindiSt,Lahore ord456 prescription.png, 2MB	Invalid Medicine ID Class	Error message for invalid medicine ID
TC_011	Invalid Prescription File Name	med123, 2 789pindiSt,Lahore ord456 prescription.draw.io, 2MB	Invalid File Name Class	Error message for invalid file name
TC_012	Invalid Prescription File Size	med123, 2 789pindiSt,Lahore ord456 prescription.png, 0MB	Invalid File Size Class	Error message for invalid file size

Chapter 5
Implementation

5 Chapter: Implementation

Below we start the implementation chapter.

5.1 Endeavor (Team + Work + Way of Working)

Team

- Usama Javed 25363
- Muhammad Talha 25362
- Muhammad Nouman 27708

5.2 Work Breakdown Structure (WBS)

- Register.
- Login.
- Edit profile.
- Chat.
- Verification.
- View lab booking.
- Accept Lab booking.
- Modify Lab Booking.
- Cancel Lab booking.
- Order Medicine.
- Upload A Prescription.
- Book Lab test.
- View Nurse Booking.
- Filter Nurse booking.
- Cancel Nurse booking.
- Complete Nurse booking.
- Payment.
- Book Nurse.
- Accept Bid.

- Cancel booking request.
- Add Lab test.
- Delete Lab test.
- Edit Lab test.
- Email Verification
- View Nurse Booking Request.
- Bid for nurse booking request.
- View Nurse Booking.
- Complete Nurse Booking.
- Cancel Nurse booking.
- Medica store change availability status.
- View medical store order request.
- View medical store pending orders.
- View medical store completed orders.
- Logout

5.3 Role and Responsibility Matrix

Table 5.1: RAM Table

WBS Name	**Activity**	**Roles (Responsible, Accountable, Consulted, Informed)**		
		M Talha	**Usama Javed**	**M Nouman**
Development	**Account Registration**	**R, A**	**C**	**I**
	Edit Profile	**R A**	**C, R**	**C, I**
	Book Lab-Test	**R, A**	**A, C**	**I, C**
	Dashboards	**C, A**	**R**	**I,**
	Verification	**R**	**A**	**C**

	View Test	**R, A**	**C**	**I**
	View Booking	**R**	**I,A**	**C**
	Order Medicine	**I,A**	**R**	**C**
	Upload A prescription	**R**	**A,C**	**I**
	Filter Nurse Booking	**R,C**	**A**	**I**
	Accept Bid	**A**	**R,I**	**C**
	Payment	**A,I**	**R**	**C**
	Edit Lab test	**R**	**AC**	**I**
	Logout	**R,A**	**C**	**I**
	Testing (Blackbox)	**R, C**	**I**	**R, A**
	Implement Database	**R, A**	**I**	**C**
	Implement Email verification	**R, A**	**R**	**C**
FYP Reports / Documentation	Write Introduction, Market Survey	**I**	**C**	**R, A**
	Maintain Functional Requirements	**I**	**R**	**C**
	Make Use Cases, Fully Dressed Use Cases	**I**	**R**	**C, A**

	Make Architecture, Design	**I**	**R, A**	**C**
	Make Test Case Designs (ECP)	**C**	**I**	**R, A**
	Maintain WBS, Roles & Responsibility, Meeting Records	**C**	**R**	**A**
WBS Name	Activity			
Training	Learn Flutter	**R, A**	**R**	**R**
	Learn how to use Tools and Technologies (MongoDB, GitHub)	**R, A**	**R, C**	**R**
Presentations	Make Project Proposal Presentation	**I**	**A**	**R**
	Make Progress Report Presentation	**I**	**R, A**	**R**

5.4 Way of Working

Muhammad Talha Developer (Mobile app-Cura-Link)

Usama Javed Documentation and App UI Design.

Muhammad Nouman Testing, Documentation.

5.5 IDE, Tools and Technologies

Now we have IDE, tools and technologies for Cura-Link.

5.5.1 Methodology

- Incremental

5.5.2 IDE

- Android Studio

5.5.3 Tools

- GitHub
- MongoDB Atlas
- Firebase
- Technologies
- Flutter
- Dart

5.5.4 Best Practices

- Software Engineering Practices
- Incremental Methodology

5.5.5 Model phases consist of SDLC

- Requirements analysis:
- Design & Development:
- Testing.
- Implementation.

5.6 Chapter Summary:

This chapter outlines the project team's composition, the Work Breakdown Structure (WBS), and each member's roles and responsibilities. It details the development tasks, training, and documentation activities, along with a RAM (Responsibility Assignment Matrix). The team adopted the Incremental development methodology using Flutter,

Dart, Android Studio, and other tools like GitHub, MongoDB, and Firebase to implement the Cura-Link app.

Chapter 6:

Testing and Evaluation

6 Chapter 6: Testing and Evaluation

Below starts the testing and evaluation chapter.

6.1 Introduction:

In this chapter you will find testing methodologies used to test our application performance and functionality along with test cases.

6.2 Test Case Design:

Now we have test cases.

6.2.1 User Registration

ECP Classes

Input (Functionality): Name, Email, Phone-no, Password

Valid Input Class:

Full name (Aa,….Zz), **Email** (Aa,…Zz)(0,1,…9)(@),

Phone-no (0,1,…,9)

Invalid Input Class:

Full name (input except valid class)**, Email** (input except valid class),

Phone-no (input except valid class).

Test Case:

Table 6.1: User Registration Test Case

Test ID	Test Title	Input (Email)	Phone-no	Full name	ECP Class	Expected Output
TC_009	Valid Signup through Email	mnkj315@gmail.com	+923155875170	Nouman khan	Valid Input Class	Sigup Successfully

TC_010	Invalid email Signup	mnkj315##.com	+923155875170	Nouman khan	Invalid input class	Error: Invalid email

6.2.2 User Login

ECP Classes

Input (Functionality): Email, Password

Valid Input Class:

Email (Aa,…Za)(0,1,…9)(@)

Invalid Input Class:

Email (input except valid class)

Test Case

Table 6.2: User Login test Case

Test ID	Test Title	Input (Email)	ECP Class	Expected Output
TC_009	Login with valid email	Mnkj315@gmail.com	Valid Input Class	Login Successfully
TC_010	Invalid email login attempt	Mnkj315##.com	Invalid Class	Invalid email format
TC_011	Email not registered	Usama123@gmail.com	Invalid Class	No Accont Exist

6.2.3 Rating & Feedback.

ECP Classes

Input (Functionality): Rating , Feedback

Valid Input Class:

Rating (1,2,3,4,5)

Feedback (valid text , blank , special character only)

Invalid Input Class:

Rating (input except valid class)

Feedback (input except valid class)

Test Case

Table 6.3: Rating & Feedback Test Case

Test ID	Test Title	Rating	Feedback	ECP Class	Expected Output
TC_001	Post valid rating & feedback	5	Perfect Service	Valid Input Class	Rating & feedback submitted
TC_002	Low Rating & Feedback	1	Not Satisfied	Valid input class	Rating & feedback submitted
TC_003	Rating 2 without feedback	2	blank	Invalid feedback class	Feedback can't be empty
TC_004	Invalid Rating	4	Over	Invalid feedback class	Feedback can't be empty

6.2.4 View Orders.

ECP Classes

Input (Functionality): Orders.

Valid Input Class:

Orders (Exist)

Invalid Input Class:

Orders (Empty)

Test Case

Table 6.4: View Order Test Case

Test ID	Test Title	Orders Present	ECP Class	Expected Output
TC_001	Post valid rating & feedback	Yes	Valid Input Class	Display list of medicine orders
TC_002	Low Rating & Feedback	No	Valid input class	Prompt: No Order Found

6.2.5 Accept Booking.

ECP Classes

Input (Functionality): Booking Status.

Valid Input Class:

Booking Status (Pending , Modification , Accepted

Invalid Input Class:

Booking Status (input except valid class)

Test Case

Table 6-1 Accept Booking Test case

Test ID	Test Title	Booking Status	ECP Class	Expected Output

TC_001	Accept after lab modification	Modified by lab	Valid Input Class	Booking status set to Accepted notify MLO
TC_002	Accept without modification	Not modified	Invalid input class	Error: Wait for lab response first

6.2.6 Modify Booking.

ECP Classes

Input (Functionality): Booking Status.

Valid Input Class:

Booking Status (Pending)

Invalid Input Class:

Booking Status (Accepted)

Test Case

Table 6.6: Modify Booking Test Case

Test ID	Test Title	Booking Status	ECP Class	Expected Output
TC_001	Modify Unaccepted lab	Pending	Valid Input Class	Booking modified successfully

TC_002	Modify accepted booking	Accepted	Invalid input class	Error: Cannot modify accepted

6.2.7 Cancel Booking

ECP Classes

Input (Functionality): Booking Status.

Valid Input Class:

Booking Status (Pending)

Invalid Input Class:

Booking Status (Accepted)

Test Case

Table 6.7: Cancel Booking test case

Test ID	Test Title	Booking Status	ECP Class	Expected Output
TC_001	Cancel unaccepted booking	Pending	Valid Input Class	Booking removed; patient and lab notify
TC_002	Cancel after booking	Accepted	Invalid input class	Error: Booking has already been accepted

6.2.8 Payment.

ECP Classes

Input (Functionality): Booking Status.

Valid Input Class:

Booking Status (Yes)

Payment Mode (valid card no)

Invalid Input Class:

Booking Status (No)

Payment Mode (invalid card no format)

Test Case

Table 6.8 Payment Test Case

Test ID	Test Title	Payment Mode	Booking Status	ECP Class	Expected Output
TC_001	Successfully payment done (lab)	4242 4242 4242 4242	Confirmed	Valid Input Class	Booking removed; patient and lab notify
TC_002	Invalid card info	4242 424 42 4	Confirmed	Invalid input class	Error: Card Number Format is not Valid

6.2.9 Add Test.

ECP Classes

Input (Functionality): Action , Test name.

Valid Input Class:

Action (Add)

Test name (New select from dropdown)

Invalid Input Class:

Action (Except add)

Test name (Duplicated)

Test Case

Table 6.9: Add Test Test case

Test ID	Test Title	Test name	Action	ECP Class	Expected Output
TC_001	Add new test	Blood Sugar	Add	Valid Input Class	Test added Successfully
TC_002	Add duplicate test	Blood Sugar	Add	Invalid input class	Error: Test already added

6.2.10 Delete Test.

ECP Classes

Input (Functionality): Action , Test name.

Valid Input Class:

Action (Delete)

Test name (Select from list)

Invalid Input Class:

Action (Except Modify)

Test name (except valid class)

Test Case

Table 6.10: Delete test Test case

Test ID	Test Title	Test name	Action	ECP Class	Expected Output
TC_001	Delete Test with confirmation	Blood Sugar	Delete	Valid Input Class	Test deleted Successfully
TC_002	Cancel Deletion	Blood Sugar	Cancel	Invalid input class	Error: Back to Test List

6.2.11 Order Medicine.

ECP Classes

Input (Functionality): Medicine . Prescription file.

Valid Input Class:

Medicine (Select or Search)

Prescription File (.Png)

Invalid Input Class:

Medicine (any special character)

Prescription File (.txt , .exe)

Test Case

Table 6.11: Order Medicine test Case

Test ID	Test Title	Test name	Prescription	ECP Class	Expected Output
TC_001	Order medicine with search medicine function	Blood Sugar	Blank	Valid Input Class	Display name Description and price of that specific medicine
TC_002	Order Medicine selecting from list	Blood Sugar	Blank	Valid input class	Display name Description and price of that specific medicine
TC_003	Order Medicine with Prescription	Blank	Mymedicicne.png	Valid input class	Redirect to offer screen where user can select once
TC_004	Order Medicine with invalid Prescription	Blank	Report .text	Invalid input class	Error: Invalid File format

6.2.12 Upload A Prescription.

ECP Classes

Input (Functionality): Prescription Format.

Valid Input Class:

Prescription File (.png)

Invalid Input Class:

Prescription File (.exe , .txt , etc)

Test Case

Table 6.12: Upload a Prescription test Case

Test ID	Test Title	Prescription File	ECP Class	Expected Output
TC_001	Valid Prescription upload	Medicine.png	Valid Input Class	Prescription uploaded successfully
TC_002	Invalid Prescription upload	Reort.txt	Invalid input class	Please Upload .png File only

6.2.13 Block Service Providers.

ECP Classes

Input (Functionality): Feedback rating , Action , Provider type

Valid Input Class:

Feedback rating Acceptable(3,4,5)

Action (No action)

Provider Type (Medical Store , Nurse , Lab)

Invalid Input Class:

Feedback rating Low(b1,2)

Action (Block)

Provider Type (Medical Store , Nurse , Lab)

Test Case

Table 6.13: Block Service providers Test case

Test ID	Test Title	Input	ECP Class	Expected Output
TC_001	Block service provider due to poor rating	Provider: Nurse Feedback: Unprofessional Rating: 1	Low Rating Class	Prescription uploaded successfully
TC_002	Acceptable Rating & Feedback no action	Provider: Lab Feedback: Excellent Rating: 5	Acceptable Rating Class	No action taken: Provider remains active

Chapter 7:

Conclusion & Outlook

7 Conclusion & Outlook

Below we start the conclusion and outlook chapter.

7.1 Introduction

This report presents the development journey of the Cura Link Project, from identifying healthcare service gap to delivering an all-in-one digital solution. The app offers features like medicine ordering, Lab test booking, and nurse hiring. All integrated with-in app chat, service bidding, and secure report sharing. Throughout the project, we applied user-centered design, tackled real world healthcare challenges, and enhanced our teamwork and technical skills.

7.2 Achievements

Below listed are our achievements.

7.2.1 Learning Aspect

Working on Cura Link greatly expanded our professional and technical skills. We enhanced our skills in flutter and explored complex modules like in app chatting, bidding system, also we applied version control through GitHub, and we learned to handle data. Preparing the final documentation improved/enhanced our communication and teamwork abilities.

7.2.2 Problems Solved

Cura Link Addresses several gaps in the healthcare services sector, particularly the fragmentation of services and lack of accessibility. Cura Link Brings together essential healthcare features like online medicine ordering, lab test booking, nurse/technician hiring into a single platform. Cura Link solves real world problems like limited access to nearby medical support, inefficient communication between service provider and patient and lack of transparency in healthcare transactions. The integrated in app chat, bidding, and report sharing features improve trust, safety, and convenience in healthcare interactions.

7.2.3 Artifacts

This project Delivers three cores:

Mobile Application, Werb Application and the last one MongoDB Altas Backend Server. These components showcase our abilities and capabilities in designing and implementing

a multi-role, real time healthcare platform. These reflect our technical growth and service as strong additions to our professional portfolios.

7.2.4 Critical Review

Although we made great effort, still there were some areas that could be improved. At time, the workload distribution was uneven, and we realized the importance of clear task management. And some early design decisions had to be revised for better performance and scalability. We encountered our challenges for features like, chat, bidding, lab verification, these all are required us to research and adapt quickly. But these experiences strengthened our problem-solving skills.

7.3 Future Recommendations

Future Recommendations below.

7.3.1 For Our Juniors Software Engineers

For those who are planning to work on similar projects for the final year project, we highly recommend forming a team with those members who are responsible, committed, and technically strong. Make a team leader who can distribute workload or task effectively, can track progress, and also maintain accountability. Use tools like Azure and WhatsApp for task management and communication. And the most important, focus on practical learning, break down problems, also make sure to communicate clearly, and use modern tools and frameworks that are commonly used in the software industry.

7.3.2 Future Plan

As Cura link has achieved its core objectives, but still several enhancements can plan for further improvement:

- Enable medical stores and labs to upload detailed service reports for further increased transparency.
- Allow image upload for test report to improve communication and reliability.
- Develop data analytics and insights to help user can track medical history and get personalized recommendations.

7.4 Summary

In conclusion, Cura Link Project Successfully Achieved key challenges in the world of healthcare systems by providing centralized, user-friendly android Application. Cura link Simplifies process of accessing medicines, booking lab tests and hiring our booking nurse/technician. Our team collaborated effectively to deliver the best user friendly and real-world Android App digital solution and gaining hands on experience in software designing and development and project management. At the end With Future upgrades, Cura Link has the Potential to involve into a reliable solution and impact solutions.

7.5 References

[1] IBM Documentation Team, "IBM Cognos Analytics Error Handling Documentation," IBM, 2022.

[2] GeeksforGeeks Editorial Team, "Software Engineering: System Design Basics," GeeksforGeeks, 2022.

[3] Guru99 Editorial Team, "Equivalence Partitioning in Software Testing," Guru99, 2022.

[4] Project Management Docs Team, "Project Planning: Work Breakdown Structure," ProjectManagementDocs.com, 2022.

[5] L. H. Trung, N. T. Binh, et al., "Transfer Learning for a Vietnamese Dialogue System," IEEE Xplore, 2022.

7.6 Reference Usage Table

Table 7-1Reference Usage Table

Ref	Used In
[1]	Throughout the document
[2]	Chapter 4 – System Design
[3]	Chapters 4 and 6 – Testing & Evaluation
[4]	Chapter 5 – Evaluation
[5]	Chapter 7 – Transfer Learning